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Final Project - Evaluating the Effect of Inference-Time CLAHE Contrast Enhancement on ResNet50 Pneumonia Detection: A Controlled Experimental Study

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1. Introduction

1.1 Background

Pneumonia is the single leading infectious cause of death in children under five and remains a persistent burden on populations worldwide (WHO, 2022). The stakes of getting the diagnosis right are high. Delayed treatment is directly associated with higher rates of complication and worse survival outcomes.

The standard first-line investigation is a chest X-ray. Pneumonia appears on a radiograph as areas of increased opacity, consolidation, or infiltrate within the lung fields that experienced radiologists can identify and characterise with reasonable reliability (Kermany et al., 2018). The difficulty is that radiological expertise is not uniformly available. That gap in human resource is one of the primary drivers of interest in automated computer-aided detection systems.

Deep learning, and convolutional neural networks (CNNs) in particular, have transformed what is achievable in automated medical image analysis. Transfer learning has become the dominant approach to chest X-ray classification (Shin et al., 2016). Models fine-tuned from architectures such as ResNet50 (He et al., 2016) have achieved performance on par with, and in some cases exceeding, individual radiologists under controlled experimental conditions (Rajpurkar et al., 2017). These results are encouraging. They do, however, carry an important caveat: they are measured on test sets drawn from the same imaging environment as the training data.

In practice, chest radiographs vary considerably in visual appearance depending on the scanner hardware, acquisition protocol, and post-processing pipeline used to produce them. Contrast is one of the most variable of these properties, and contrast variation across imaging environments is a recognised source of performance degradation when models encounter images that look different from what they were trained on. One practical response is to normalise image contrast before it reaches the model at the input stage rather than by retraining. Contrast Limited Adaptive Histogram Equalisation (CLAHE) is a well-established image processing technique that redistributes pixel intensities within local image regions to recover structural detail suppressed in low-contrast areas, while a clip limit prevents noise amplification in homogeneous regions (Zuiderveld, 1994). It has been applied as a preprocessing step in several chest X-ray classification studies and reported to improve model performance when applied uniformly across training and test data (Diana et al., 2025).

What has not been investigated is how CLAHE behaves when applied exclusively at inference time that is, when the model was trained on unprocessed images and CLAHE is introduced only during evaluation. A deployed model is often fixed yet incoming images may arrive with contrast characteristics that differ from the training distribution. Whether applying CLAHE to those images at the point of prediction improves, degrades, or leaves unchanged the model's performance is an empirical question that has not been systematically tested. That is the question this project addresses.

1.2 Aims and Objectives

The aim of this project is to design, implement, and evaluate a controlled experimental pipeline that assesses whether CLAHE contrast enhancement applied exclusively at inference time improves ResNet50 pneumonia detection performance, and how that effect varies across different CLAHE parameterisations.

This aim is pursued through the following objectives:

1. Obtain and preprocess the Kermany et al. (2018) chest X-ray dataset, constructing a stratified 70/15/15 train, validation, and test partition that preserves the natural class distribution.
2. Conduct exploratory pixel-level analysis of the dataset to characterise the baseline intensity and contrast properties of normal and pneumonia images, and to quantify the proportion of low-contrast cases.
3. Fine-tune a ResNet50 architecture, pretrained on ImageNet, on the unprocessed training subset using three fine-tuning strategies using head only, partial unfreezing of the final residual block, and full network fine-tuning and select the best-performing configuration based on AUC.
4. Evaluate the trained model on the held-out test set under four CLAHE conditions: a baseline with no preprocessing and three parameterised conditions varying clip limit and tile grid size, computing AUC, sensitivity, specificity, F1, and accuracy per condition.
5. Apply bootstrap confidence intervals, DeLong AUC comparison tests, and McNemar paired outcome tests to confirm the statistical significance of observed performance differences between conditions.
6. Apply Grad-CAM to representative true positive, false negative, and false positive cases under each condition to provide visual, mechanistic evidence for the observed performance patterns and to assess whether CLAHE changes the spatial focus of model attention.

1.3 Scope of the Study

This study investigates binary classification of pneumonia from chest radiographs using a single publicly available dataset. The experimental scope is centred on one specific question: does applying CLAHE preprocessing exclusively at inference time, to a model trained on unprocessed images, improve classification performance, and does the answer depend on the parameterisation chosen?

The investigation is conducted at the level of image input manipulation rather than architectural modification or model retraining. It does not involve cross-site data collection or real-world deployment testing. The experimental design evaluates CLAHE as a controlled parameter sweep within a single dataset rather than a simulation of specific institutional contrast profiles. Findings should be interpreted as a controlled, proof-of-concept

contribution to the question of inference-time preprocessing, situated within a supervised deep learning framework using a publicly available benchmark.

1.4 Contribution of this Study

The primary contribution of this project is a controlled, statistically rigorous evaluation of CLAHE preprocessing applied exclusively at inference time on a ResNet50 pneumonia detection model. Unlike existing studies, which apply contrast enhancement uniformly across training and evaluation, this study isolates the effect of inference-time CLAHE on a model that has never seen enhanced images during training. This design directly tests whether CLAHE functions as a useful deployment-time intervention and, importantly, identifies the conditions under which it may cause harm rather than benefit.

The secondary contribution is a Grad-CAM interpretability analysis that provides mechanistic, spatially grounded evidence for the observed performance patterns, going beyond aggregate metrics to explain why each CLAHE condition produces the results it does.

2. Literature Review

2.1 Deep Learning for Chest X-Ray Classification

The application of CNNs to chest radiographic analysis has been one of the most extensively studied areas in medical image classification over the past decade. The benchmark that dominates this literature is the Kermany et al. (2018) dataset: 5,863 labelled paediatric chest X-ray images from Guangzhou Women and Children's Medical Centre, split into pneumonia and normal categories. Its clean binary labelling and unrestricted public availability have made it the standard training resource for binary pneumonia detection. A systematic review of thirteen deep learning studies on chest X-ray pneumonia detection confirmed that ten of the thirteen papers used this dataset (Pratama et al., 2023), that makes it both a useful benchmark and a potential source of overfitting to a single institutional imaging profile.

Within-dataset performance figures are consistently high in the literature, with reported accuracy values averaging above 90% and reaching as high as 99.62% in individual studies (Pratama et al., 2023). These figures are measured on held-out test sets drawn from the same distribution as training data. The same systematic review established that sensitivity is the most clinically critical metric in this domain, because a false negative, a pneumonia case classified as normal, poses substantially greater patient risk than a false positive since missed diagnosis leaves the condition untreated (Pratama et al., 2023). This prioritisation directly informs the evaluation framework adopted in the present study, where sensitivity and specificity are reported alongside AUC as primary metrics, with sensitivity given particular weight. The dataset, however, is not class balanced. Approximately 73% of images are pneumonia cases and 27% normal, which can inflate recall at the expense of specificity if left unaddressed. As per Alshanketi et al. (2025) combining class weighting with data augmentation improved accuracy on this dataset from 81% to 91%. So, the study applies normalised class weights computed from training set proportions, assigning a weight of 2.70 to the minority normal class.

2.2 Transfer Learning in Medical Imaging

Transfer learning has become the standard approach to medical image classification when labelled data is limited. Early convolutional layers of networks trained on natural images learn general low-level representations including edges, textures, and orientations that transfer effectively across domains, while deeper layers become progressively specialised to the source task and may require fine-tuning or replacement (Yosinski et al., 2014). This layered transferability provides a principled justification for initialising medical image classifiers from ImageNet-pretrained weights even when the source and target domains differ as substantially as natural photographs and greyscale radiographs. ResNet50 has emerged as one of the most widely adopted backbone architectures for this purpose. Its residual connections help with the vanishing gradient problem that limited very deep network training, allowing gradients to propagate reliably through 50 layers (He et al., 2016). Diana et al. (2025) compared ResNet50 against DenseNet, EfficientNet, and MobileNetV2 and found ResNet50 came out on top across every preprocessing condition they tested. Zunaed et al. (2024) independently made the same choice for a cross-domain pneumonia detection

framework. It was also chosen for a cross-domain pneumonia detection framework (Zunaed et al., 2024). Hence, supporting its selection as the backbone for this study.

A recognised limitation of applying ImageNet-pretrained models to chest radiography is the modality gap that ImageNet consists of colour photographs of natural scenes, while chest X-rays are greyscale projections of internal anatomy. Raghu et al. (2019) examined this gap empirically and found that ImageNet-pretrained initialisations still outperform random initialisation for medical imaging tasks, particularly when the target dataset is of moderate size. This supports the use of transfer learning here and suggests that pretrained representations provide a useful prior for pulmonary structure recognition despite the domain difference.

The extent of fine-tuning is a design decision with meaningful performance consequences. Training only the classification head freezes the backbone entirely, minimising the risk of catastrophic forgetting of pretrained representations. Unfreezing deeper layers allows adaptation to the target domain but introduces overfitting risk, particularly when the target dataset is small relative to the number of parameters being updated (Shin et al., 2016). The present study evaluates three strategies with head only, partial unfreezing of the final residual block, and full network fine-tuning to determine empirically which best balances domain adaptation against representation preservation.

2.3 Contrast Enhancement using CLAHE

Contrast variation is a practically significant source of image-level difference in chest radiography, arising from differences in scanner hardware, acquisition protocols, and post-processing pipelines across clinical sites and imaging conditions. Standard global histogram equalisation redistributes pixel intensities to maximise contrast across the full image but produces unreliable results on chest radiographs where lung fields, mediastinal structures, and diaphragm have substantially different local intensity profiles (Gonzalez and Woods, 2018). Adaptive Histogram Equalisation (AHE) addresses this by computing separate local histograms for different image regions but introduces noise amplification in areas of low local variance. Contrast Limited Adaptive Histogram Equalisation (CLAHE) was developed to resolve this by applying a clip limit to each local histogram before redistribution, suppressing noise amplification while preserving local contrast enhancement (Zuiderveld, 1994). It has been applied successfully in medical imaging more broadly, including in enhancement of low-contrast portal films for which it was originally developed.

Several studies have applied CLAHE as a preprocessing step in chest X-ray classification pipelines and reported performance improvements. The most relevant comparison point is Diana et al. (2025). They ran a sequential pipeline — gamma correction first, then CLAHE with clipLimit = 2.0 and tileGridSize = (8,8) — across chest X-ray data from three Indonesian hospital sources.. ResNet50 with this preprocessing achieved 89.12% testing accuracy under 10-fold cross-validation, compared to 83.65% without preprocessing which is a gain of over 5 percentage points. The study did not evaluate gamma correction and CLAHE in isolation, so their individual contributions cannot be separated. Despite this, the parameterisation provides the most rigorously evaluated CLAHE configuration for chest X-ray ResNet50 classification in the recent literature, and is adopted as condition C1 in the

present study. Singthongchai and Wangkhamhan (2023) investigated CLAHE as a preprocessing intervention and reported improved classification performance relative to an unprocessed baseline in chest X-ray classification tasks.

A critical observation amongst all these studies is that CLAHE is applied uniformly across both training and test data. The model is trained on enhanced images and evaluated on enhanced images. This is internally consistent, but it does not answer a different and practically important question: what happens when CLAHE is applied only at inference time, to a model trained without it? This inference-time configuration presents a scenario common in deployment that is, a model is fixed and cannot be retrained, but incoming images may benefit from contrast normalisation. No existing study in this literature has tested this configuration in a controlled way. This study fills that gap by training ResNet50 on unprocessed images and evaluating it under four systematically varied CLAHE conditions, with full statistical significance testing.

2.4 Model Interpretability using Grad-CAM

Strong quantitative performance is insufficient evidence of clinical reliability on its own. A model that achieves high accuracy may do so by attending to positioning markers, acquisition artefacts, or other non-pathological features rather than the tissue abnormalities a clinician would interpret. A survey of explainability methods for medical AI identified the need for spatial validation confirming not just that a model performs well, but that it does so for anatomically coherent reasons (Tjoa and Guan, 2020). Gradient-weighted Class Activation Mapping (Grad-CAM) is a post-hoc visualisation method that generates spatial heatmaps showing which regions of an input image most influenced a CNN's prediction (Selvaraju et al., 2017). It computes the gradient of the class score with respect to the final convolutional layer's feature maps, averages these gradients spatially to produce per-channel importance weights, and generates a weighted sum that is upsampled and overlaid on the input image. Grad-CAM can be applied directly to any pretrained CNN. It was found that correctly classified pneumonia cases showed attention concentrated over opacified lung regions, while misclassified cases showed displaced or diffuse attention when GRAD-CAM was used for interpretation (Ennab and Mcheick, 2022).

The application of Grad-CAM in the present study goes beyond standard interpretability validation. By applying it across all four CLAHE conditions on the same images, the analysis can directly examine whether and how preprocessing changes the model's spatial attention. A condition that improves sensitivity while maintaining coherent lung-field attention tells a different story from one that improves sensitivity while producing gradient collapse. This condition-comparative use of Grad-CAM, applied systematically across experimental conditions, provides mechanistic evidence beyond aggregate metrics.

2.5 Research Gap

Three things are clear, deep learning models fine-tuned from ImageNet-pretrained ResNet50 are capable of strong pneumonia detection performance on the Kermany dataset; CLAHE is an effective contrast enhancement technique that improves chest X-ray classification performance when applied consistently across training and evaluation, and Grad-CAM provides a practical and non-invasive tool for spatial validation of model attention that can be used to interpret performance patterns beyond aggregate metrics.

What none of this literature addresses are the inference-time configuration: applying CLAHE to test images from a model trained without it. When CLAHE is applied only at inference time, the model encounters an input distribution it has never seen during training. Whether this improves, degrades, or leaves unchanged the model's classification performance and whether any change is driven by genuine improvement in feature detection or by an artefactual shift in the input distribution are open empirical questions. The study is, to the best of the author's knowledge, the first to investigate this configuration in a controlled, statistically rigorous manner on a standard chest X-ray benchmark, with Grad-CAM used to provide mechanistic interpretation of the observed performance patterns.

3. Methodology

This chapter presents the full experimental methodology for evaluating the effect of CLAHE contrast enhancement preprocessing on ResNet50 transfer learning performance for binary pneumonia detection. The experimental design follows two stages to isolate effects cleanly: a model-selection stage, in which the optimal fine-tuning strategy is identified using validation performance only, and a preprocessing-effect stage, in which the selected model is fixed and evaluated under four preprocessing conditions on the held-out test set. This structure prevents confounding between architecture-selection effects and preprocessing effects. An overview of the pipeline is provided in Figure 3.1.

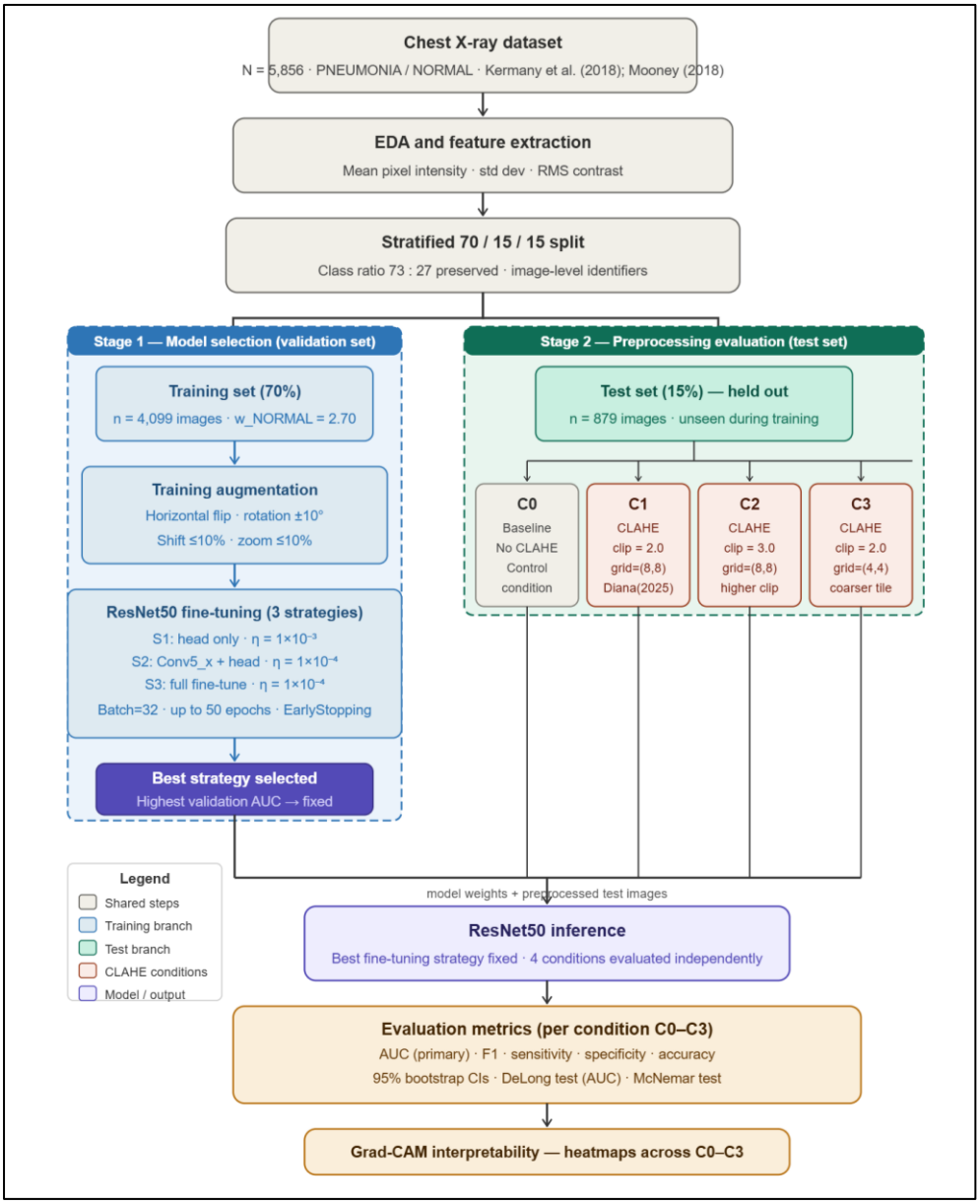


Figure 3.1 - End-to-end experimental pipeline.

3.1 Software and Computational Environment

All experiments were implemented in Python 3.10 and executed on Google Colab Pro, using an NVIDIA T4 GPU with a high-RAM runtime. The complete set of software dependencies are listed in Table 3.1.

Tool/Library	Version
Python	3.10
Google Colab Pro	
TensorFlow / Keras	2.16.1
OpenCV	4.12.0
Scikit-image	0.26.0
NumPy	2.0
Pandas	2.2.2
Matplotlib / Seaborn	3.10.8
Scikit-learn	1.8.0
SciPy	1.17.1

Table 3.1 - Software/Libraries used

3.2 Dataset

The dataset used in this study is the Chest X-Ray Images (Pneumonia) collection, originally curated by Kermany et al. (2018) and made publicly available on Kaggle by Paul Mooney (Mooney, 2018), containing 5,856 anterior-posterior chest radiographs from paediatric patients at the Guangzhou Women and Children's Medical Centre, China. Images are distributed across two diagnostic classes: Pneumonia (n = 4,273; 73.0%) and Normal (n = 1,583; 27.0%). It is noted that the original publication reports a total of 5,863 images; the discrepancy is attributed to missing files in the Kaggle-hosted version and is recorded as a methodological limitation.

3.2.1 Class Imbalance and Weighting

The dataset exhibits a class imbalance of approximately 2.70 : 1 (pneumonia to normal). This imbalance reflects the genuine clinical prevalence of pneumonia in the source population, but it creates a problem during training that a model can achieve deceptively high accuracy by simply predicting pneumonia for most inputs. To counteract this, class weights were calculated from the training set distribution using *sklearn.utils.class_weight.compute_class_weight*, with the normal class assigned a weight of 2.70 relative to pneumonia. This weight is applied as a scaling factor to the loss function during training.

Balanced class weights are computed as follows:

$$w_c^{bal} = \frac{N_{train}}{(K \times n_c)}$$

(Eq. 3.1)

where N_{train} is the total number of training samples, K is the number of classes, n_c is the number of training samples belonging to the specific class and w_c^{bal} is the resulting weight assigned to class c during training. Weights are normalised by dividing by the smallest raw weight, so that the majority class receives a weight of 1.0.

3.2.2 Dataset Splitting

The dataset was partitioned into training, validation, and test sets using a stratified 70/15/15 split implemented via *sklearn.model_selection.train_test_split* with the stratify parameter set to the class label vector, preserving the 73:27 class ratio across all three partitions. As only image-level identifiers are available in the Kaggle release, partitioning was performed at image level; the absence of patient-level identifiers is recorded as a methodological limitation that may introduce a small risk of data leakage if the same patient appears across splits. The resulting partition is summarised in Table 3.2 and illustrated in Figure 3.2.

Split	Total Images	Pneumonia	Normal	Pneumonia: Normal Ratio
Train (70%)	4099	2991 (73%)	1108 (27%)	2.70 : 1
Validation (15%)	878	641 (73%)	237 (27%)	2.70 : 1
Test (15%)	879	641 (72.9%)	238 (27%)	2.69 : 1

Table 3.2 - Dataset partition summary

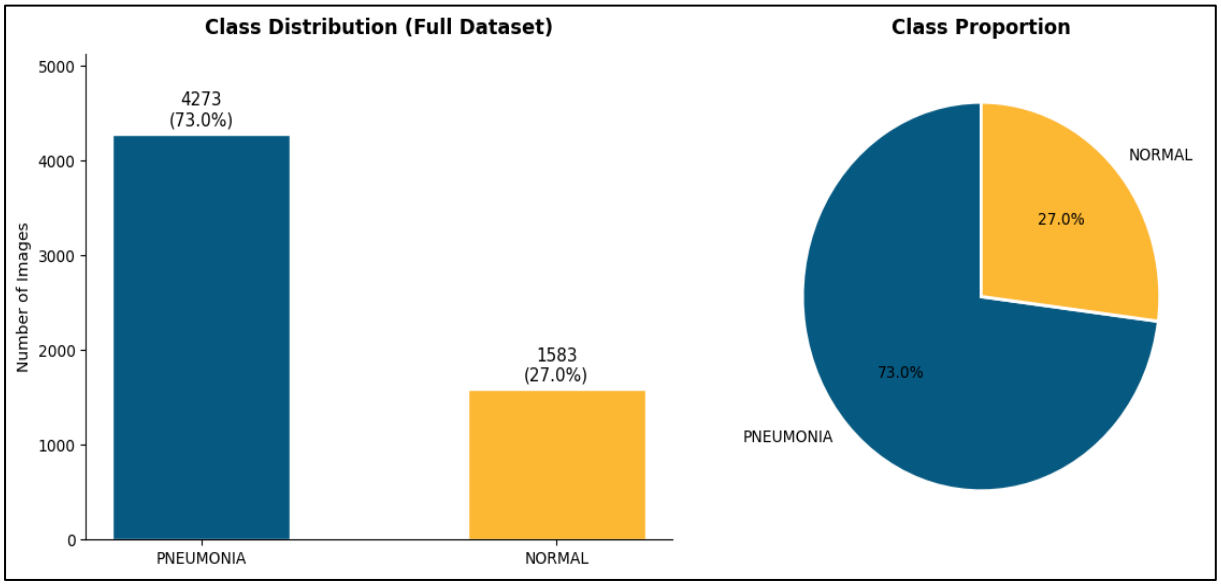


Figure 3.2 - Class distribution across dataset splits

3.2.3 Image-Level Statistical Characteristics

Exploratory analysis was conducted on a representative sample to characterise the baseline pixel-level properties of the dataset prior to preprocessing, with full analysis and confirmed statistics presented in Section 4.1. Preliminary findings indicated that mean pixel intensity showed no significant discriminative value between classes, while the standard deviation of pixel intensity was higher for normal images than pneumonia images, consistent with the clinical expectation that pulmonary consolidation homogenises lung tissue appearance and reduces local intensity variation (Walker et al., 2014). RMS contrast analysis further revealed a pronounced low-contrast tail in pneumonia images that was largely absent from the normal distribution (Figure 3.4), indicating a subset of cases with suppressed contrast where discriminative features may be difficult for the model to detect. This observation directly motivates the use of CLAHE as a preprocessing intervention, the visual effect of which across all four conditions is illustrated in Figure 3.5 (Zuiderveld, 1994).

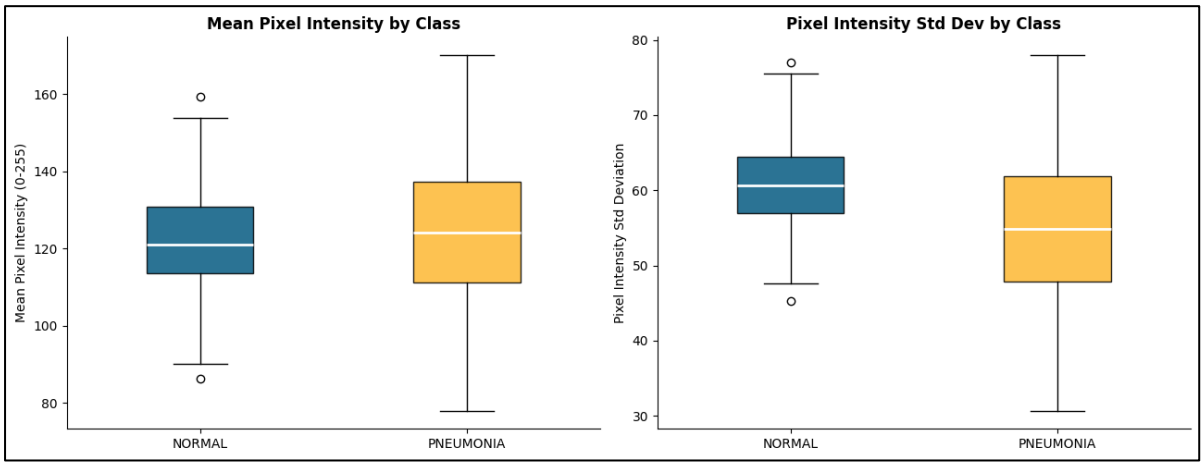


Figure 3.3 - Baseline pixel intensity statistics by class. Normal images exhibit marginally higher intensity standard deviation

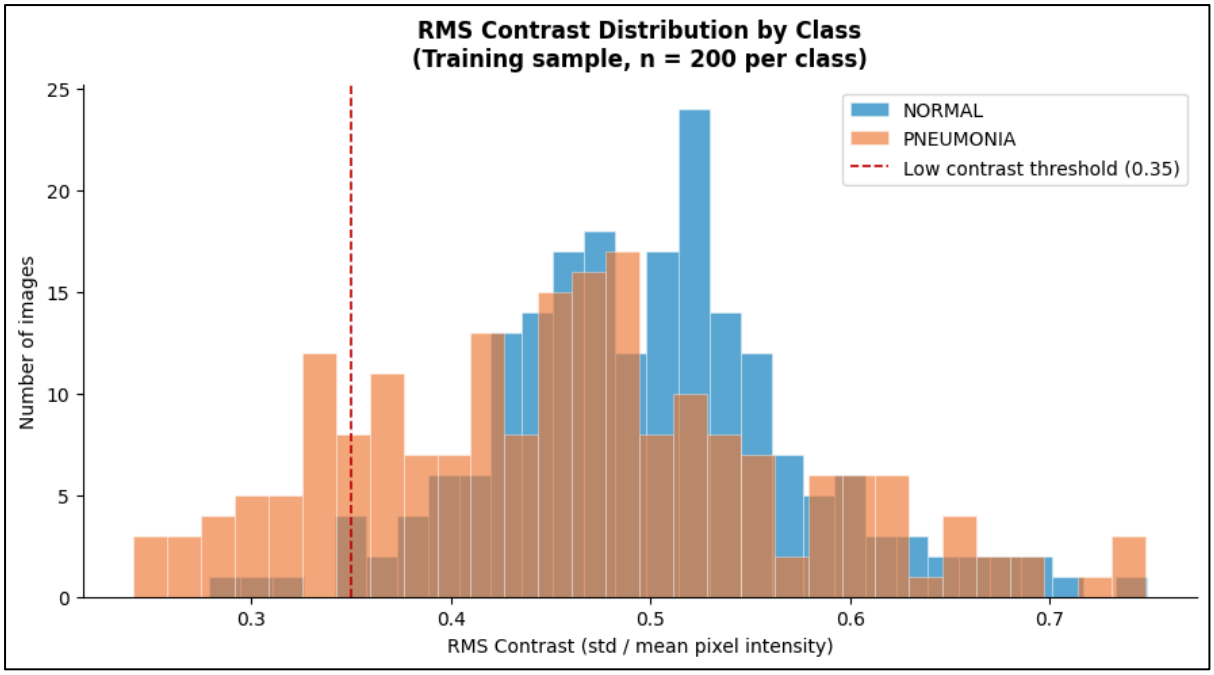


Figure 3.4 - RMS contrast distribution by class. The low-contrast tail in Pneumonia (0.20–0.35) observed.

3.3 Image Preprocessing

All images were converted to greyscale prior to preprocessing, as all clinically meaningful structural information in a chest radiograph is conveyed through pixel intensity values, which reflect the differential attenuation of X-rays across tissues of varying density. Images were subsequently resized to 224 × 224 pixels to match the input dimensions required by the ResNet50 architecture. Pixel values were then normalised using the ResNet50 ImageNet normalisation convention, subtracting the channel-wise means [103.939, 116.779, 123.68] from each channel. This aligns the input distribution with what the pretrained weights expect and is a requirement for meaningful transfer from ImageNet-pretrained initialisations. The resulting value range after normalisation was confirmed as approximately [−123.68, 150.79]. These steps were applied uniformly across all conditions.

3.3.1 Training Image Augmentation

To improve model generalisation and reduce overfitting given the fixed training set size, a set of random augmentations was applied to training images at load time using the TensorFlow *tf.keras.preprocessing.image.ImageDataGenerator* pipeline. The following transformations were applied with randomised parameters on each batch: horizontal flip, rotation (up to ±10°), width and height shift (up to 10%), and zoom (up to 10%). The transformations were chosen to simulate the natural variation in patient positioning and image acquisition that occurs in clinical practice. Augmentation was applied to the training set only; validation and test images were not augmented.

3.3.2 CLAHE Preprocessing

Unlike global histogram equalisation, which applies a single intensity mapping across the entire image, CLAHE partitions the image into a grid of non-overlapping contextual tiles and applies histogram equalisation independently within each tile, with bilinear interpolation at tile boundaries to eliminate tiling artefacts. Within each tile, the histogram is clipped and redistributed in two steps. First, any histogram bin whose count exceeds the predefined clip limit is reduced to that limit value. Pixels cut from the overrepresented bins do not disappear. They get redistributed across the other bins in equal portions, keeping the overall pixel count for that tile intact.

After redistribution, the bins are accumulated from left to right to produce a mapping. Each input intensity value is then assigned a new output value based on where it sits in this mapping, which has the effect of pulling the intensities apart and making the local contrast more visible. For each pixel in tile t, the new output intensity is computed as:

$$s_t(r) = \frac{L-1}{N_t} \sum_{i=0}^{r} \tilde{h}_t(i)$$

(Eq. 3.2)

where $s_{t(r)}$ is the output intensity for a pixel with input intensity r in tile t, L is the number of intensity levels (256), N_t is the total number of pixels in tile t, and $h_t(i)$ is the redistributed histogram count at bin i. By constraining the histogram before this mapping is applied, the clip limit limits how steeply the cumulative distribution function can rise at any point, which in turn bounds the degree of contrast enhancement and prevents noise from being excessively amplified (Zuiderveld, 1994). The final enhanced image is produced by applying this tile level mapping across the image and interpolating smoothly between adjacent tile boundaries to avoid visible block artefacts.

3.3.3 Experimental CLAHE Conditions

Four test-time preprocessing conditions are evaluated, defined in Table 3.3. CLAHE is applied to test set images only; the model is trained once on unprocessed images per fine-tuning strategy. This isolates the preprocessing effect at inference time, keeping the trained model constant across all four conditions so that any observed performance differences are attributable solely to the preprocessing applied.

Condition	Clip Limit (C_L)	Grid Size	Description
C_0	N/A	N/A	Baseline; No pre-processing beyond resize and normalisation
C_1	2.0	(8, 8)	Standard CLAHE
C_2	3.0	(8, 8)	Higher clip limit with stronger enhancement
C_3	2.0	(4, 4)	Finer tile grid with 16 tiles rather than 64, meaning each tile covers a larger image region and adaptation is less spatially localised than C1

Table 3.3 - Experimental CLAHE conditions (C0–C3) evaluated on the test set.

Comparing C1 against C2 isolates the effect of clip limit (2.0 vs 3.0) at fixed tile size. Comparing C1 against C3 isolates the effect of tile grid size (8×8 vs 4×4) at fixed clip limit. This factorial structure allows the independent contribution of each CLAHE parameter to be assessed.

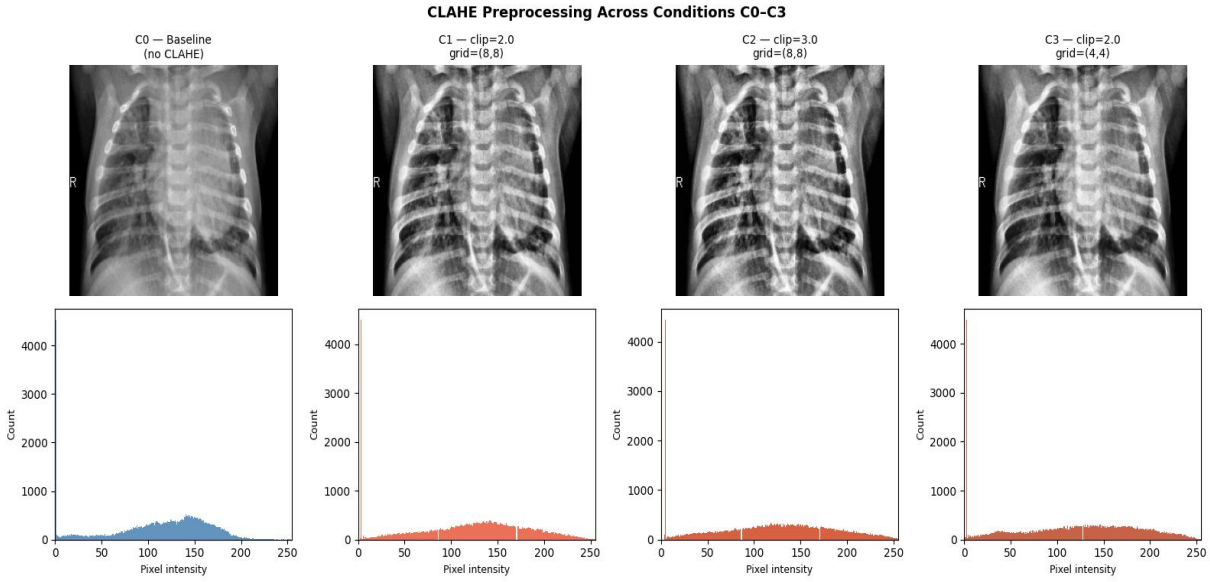


Figure 3.5 - Visual and histogram comparison of the four preprocessing conditions applied to a representative chest radiograph.

3.4 Model Architecture of ResNet50 with Transfer Learning

A ResNet50 convolutional neural network, pre-trained on the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) dataset, was adopted as the classification backbone. Deep networks commonly struggle during training because the signals used to update weights tend to fade before they reach the earlier layers. ResNet gets around this by adding the block input directly onto its output, so the update signal always has a direct path back regardless of depth (He et al., 2016).

3.4.1 Residual Learning

In a standard network layer, the input goes in, gets transformed, and the result comes out. ResNet adds one change to this: the original input is also added onto the output before it moves to the next layer. In practice this means each block only needs to figure out what to add or adjust. Each block learns the residual mapping *F(x)* such that the overall block output is:

$$H(x) = F(x) + x$$

(Eq. 3.3)

where H(x) is the desired block output, F(x) is the residual function learned by the stacked weight layers, and x is the block input passed directly through the shortcut connection. If the identity mapping is already optimal, the network drives F(x) toward zero, making optimisation easier than learning the full mapping from scratch. Gradients flow directly through the shortcut path during backpropagation, mitigating vanishing gradients at depth (He et al., 2016). Each residual block in ResNet50 adopts a bottleneck configuration consisting of three convolutions. The full architecture is summarised in table 3.4.

Stage	Component
Conv1 + Pool	Initial feature extraction using a 7×7 convolution with stride 2 to halve spatial dimensions, followed by a 3×3 max pooling layer with stride 2 for further down sampling
Conv2x	3 repeated bottleneck residual blocks, each consisting of 1×1, 3×3, and 1×1 convolutions with output depth of 256 channels
Conv3x	4 repeated bottleneck residual blocks with increased feature depth, producing 512 channel feature maps at half the spatial resolution of Conv2x
Conv4x	6 repeated bottleneck residual blocks representing the deepest stage of feature abstraction, producing 1024 channel feature maps
Conv5x	3 repeated bottleneck residual blocks producing the final convolutional feature maps at 2048 channels; this stage is unfrozen in fine-tuning strategy S2
Global Average Pool	Spatial averaging across the full 7×7 feature map, reducing each of the 2048 channels to a single value and producing a compact 2048-dimensional feature vector
Custom Head	Fully connected layer of 256 units with ReLU activation for classification learning, followed by dropout at rate 0.5 for regularisation, and a final 2-unit SoftMax output layer producing class probabilities for normal and pneumonia

Table 3.4 - ResNet50 architecture components

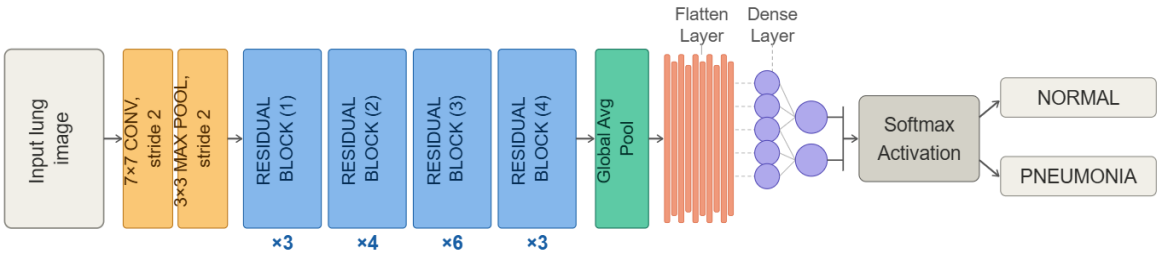


Figure 3.6 - ResNet50 bottleneck residual block

3.4.2 Fine-Tuning Strategies

Three strategies were compared rather than adopting a single approach because the optimal degree of fine-tuning for medical image classification tasks is not known in advance and varies depending on how much the target domain differs from the ImageNet source domain and how much labelled training data is available (Shin et al., 2016; Raghu et al., 2019). Evaluating all three strategies on the validation set ensures the final model selection is empirically justified rather than assumed, defined in Table 3.5. Each strategy was trained independently on the same training set using the same hyperparameters and evaluated on the validation set. The strategy achieving the highest validation AUC was then selected as the fixed model for the CLAHE preprocessing comparison experiment (Section 3.3.3). This two-stage design ensures that architecture choices and preprocessing effects are evaluated separately, preventing confounding between the two experimental variables.

Strategy	Layers Trained
S1	Custom head only (Dense layer + Dropout + SoftMax)
S2	Conv5x Residual block + custom head
S3	All layers (full fine-tune)

Table 3.5 - Three fine-tuning strategies evaluated in Stage 1

3.5 Training Configuration

All three strategies were trained using the Adam optimiser with a batch size of 32 and a maximum of 50 epochs. Strategy S1 was trained with a learning rate of 1×10^{-3}, while strategies S2 and S3 were trained at a lower rate of 1×10^{-4} to preserve the pretrained ImageNet representations. The loss function used during training was weighted categorical cross-entropy, which penalises misclassification of the minority class more heavily through the application of class weights. This is expressed as:

$$\mathcal{L} = -\frac{1}{N_b}\sum_{i=1}^{N_b}\sum_{c=1}^{2} w_c \cdot y_{i,c} \cdot \log(\hat{y}_{i,c})$$

(Eq. 3.4)

where N_b is the batch size, $y_{\{i,c\}}$ is the ground truth label, $\hat{y}_{i,c}$ is the predicted class probability, and w_c is the class weight applied to class c, with w_{Normal} = 2.70 and $w_{Pneumonia}$ = 1.0. Three callbacks were applied throughout training. Early stopping watched validation AUC and cut training short if it had not improved in 10 consecutive epochs, preventing the model from continuing to update once the useful learning had stopped. A learning rate scheduler stepped in after 5 epochs without progress, halving the rate to allow finer adjustments near the optimum. Separately, a checkpoint was saved each time validation AUC reached a new best so regardless of when training ended, the weights from the strongest epoch were always the ones taken forward to evaluation.

3.6 Evaluation Metrics

Final evaluation was conducted on the held-out test set (n = 879) under each of the four preprocessing conditions C0 to C3, using the best performing fine-tuning strategy identified in Stage 1. Five metrics were computed for each condition. Area under the ROC curve (AUC) was adopted as the primary metric as it evaluates the model's ability to discriminate between classes across all decision thresholds, making it robust to class imbalance, which can be mathematically expressed as:

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}\left(\mathrm{FPR}^{-1}(u)\right) du$$

(Eq. 3.5)

The remaining four metrics were computed at a fixed decision threshold of $\tau = 0.5$:

$$Accuracy = \frac{(TP+TN)}{(TP+TN+FP+FN)}$$

(Eq. 3.6)

$$Sensitivity = \frac{TP}{(TP+FN)}$$

(Eq. 3.7)

$$Specificity = \frac{TN}{(TN+FP)}$$

(Eq. 3.8)

$$F1 = 2 \times \frac{(\mathrm{Pr}ecision \times Sensitivity)}{(\mathrm{Pr}ecision+Sensitivity)}$$

(Eq. 3.9)

where TP is the number of true positives representing pneumonia cases correctly identified, TN is the number of true negatives representing normal cases correctly identified, FP is the number of false positives representing normal cases incorrectly classified as pneumonia, and FN is the number of false negatives representing pneumonia cases incorrectly classified as normal. Sensitivity is given interpretive weight in this study as a false negative, where a pneumonia case goes undetected, carries a greater clinical cost than a false positive. Rather than reporting each metric as a single number, 95% bootstrap confidence intervals were computed over 1,000 resamples to give a sense of how stable each estimate is. Two statistical tests were then used for pairwise condition comparisons, each chosen because of how well it fits this experimental setup.

Pairwise comparisons between conditions were conducted using two statistical tests selected for their suitability to this experimental design. The DeLong et al. (1988) test was used to compare AUC values between conditions. Because all four conditions were evaluated on the same 879 test images, the AUC estimates are correlated — a standard independent samples test would produce artificially inflated z-statistics by ignoring this shared structure. The DeLong test accounts for this correlation through the structural component approach, producing valid z-statistics and two-tailed p-values for each pairwise comparison. The McNemar test was used to compare the pattern of binary classification outcomes between conditions, testing whether the shift in which specific images were correctly or incorrectly classified was statistically significant rather than driven by chance. A continuity correction was applied throughout. Both tests used a significance threshold of $\alpha = 0.05$.

4. Implementation and Results

4.1 Exploratory Pixel Analysis

Exploratory analysis was conducted on 200 training images per class. Mean pixel intensity showed no significant discriminative value between classes (normal median 124.2, pneumonia 123.3 i.e. a difference of less than 1 unit on a 0 to 255 scale). Pixel intensity standard deviation was significantly higher in normal images (median 61.4, IQR 57.0 to 64.7) than pneumonia images (median 53.6, IQR 47.9 to 63.8), consistent with consolidation homogenising lung tissue and reducing local intensity variation (Mann-Whitney $U = 27{,}531$, $p < 0.001$; Walker et al., 2014). Critically, 18.5% of pneumonia images fell below an RMS contrast of 0.35, compared to 2.5% of normal images ($U = 24{,}992$, $p < 0.001$) identifying a substantial low-contrast pneumonia subset where discriminative features are suppressed. This directly motivates contrast enhancement as a preprocessing intervention. Figure 4.2 and Figure 4.3 illustrates the CLAHE preprocessing effect across all four conditions.

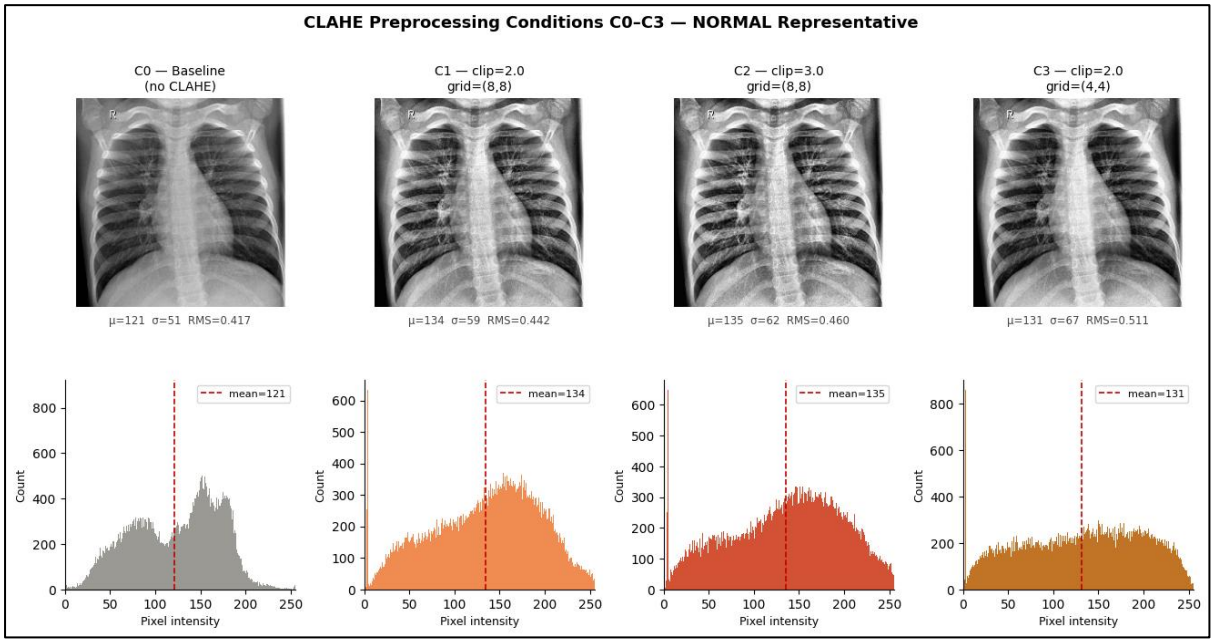


Figure 4.2 - Visual appearance and pixel intensity histograms for conditions C0 through C3 applied to a normal image.

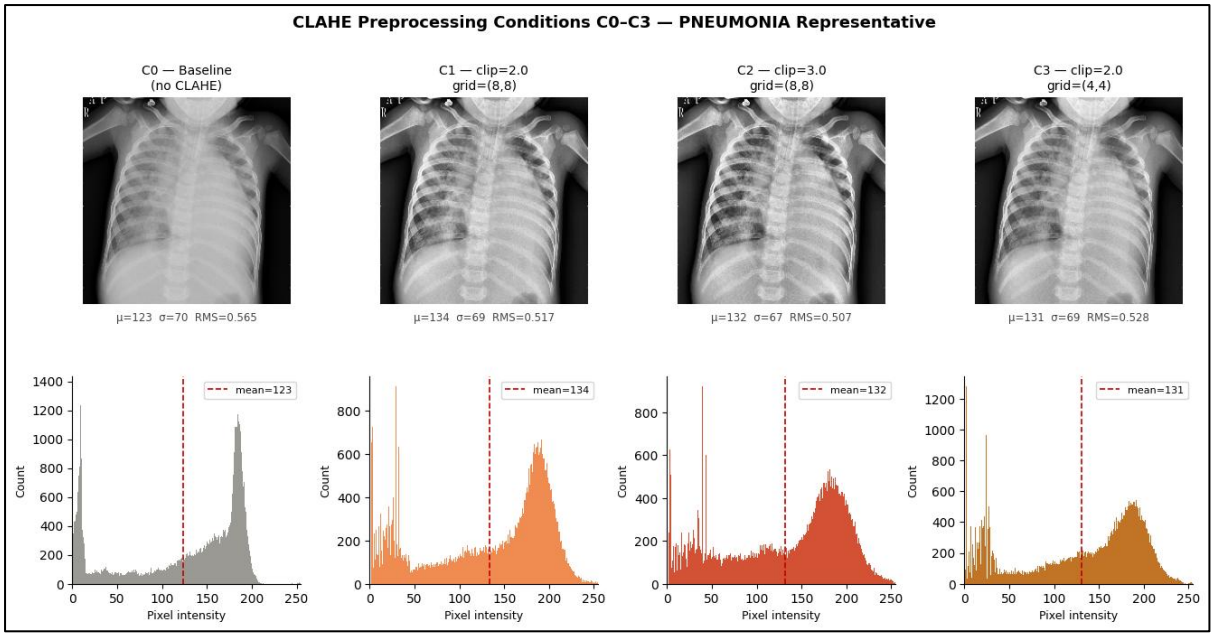


Figure 4.3 - Visual appearance and pixel intensity histograms for conditions C0 through C3 applied to a pneumonia image.

4.2 Experimental Setup

The primary artefact is a fourteen-block Jupyter notebook implemented on Google Colab Pro with an NVIDIA T4 GPU, fully self-contained and reproducible from Block 1. All image transformations used a single unified preprocessing function across all conditions, CLAHE was applied conditionally, with C0 passing images through unmodified. Images were loaded in greyscale, resized to 224 × 224, channel-replicated to satisfy ResNet50's three-channel input requirement, and normalised using the ResNet50 ImageNet convention (subtracting channel-wise means; confirmed output range [−123.68, 151.05]). A custom *tf.keras.utils.Sequence* generator was implemented to enable switching between CLAHE conditions at inference time without any other change to the pipeline. Training augmentation (horizontal flip, ±10° rotation, ±10% shift, 0.9–1.1 zoom) was applied to the training set only. Pipeline integrity was confirmed via batch shape (32, 224, 224, 3), dtype, and value range verification before training began.

ResNet50 was loaded with pretrained ImageNet weights, with the original head replaced by global average pooling, a 256-unit dense layer with ReLU, dropout at 0.5, and a 2-unit softmax output. The full architecture contains 24,112,770 parameters. All three fine-tuning strategies used Adam, batch size 32, maximum 50 epochs, and weighted categorical cross-entropy with $w_{NORMAL} = 2.70$ and $w_{PNEUMONIA} = 1.0$. Three callbacks were applied uniformly: early stopping (patience 10), learning rate halving on plateau (patience 5), and model checkpointing on best validation AUC. Strategy S1 trained 525,058 parameters (2.2%), S2 trained 15,501,058 (64.3%), and S3 trained 24,059,650 (99.8%).

4.3 Stage 1: Fine-Tuning Strategy Comparison

Three fine-tuning strategies were trained from identical ImageNet-pretrained weights with global seeds reset before each build. Table 4.1 summarises validation performance and Figure 4.4 shows training curves.

Strategy	AUC	Accuracy	Sensitivity	Specificity	Epochs
S1	0.9848	0.9510	0.9750	0.8861	12
S2	0.9913	0.9624	0.9719	0.9367	11
S3	0.9950	0.9795	0.9844	0.9662	22

Table 4.1 - Stage 1 validation performance across fine-tuning strategies (best weights after early stopping).

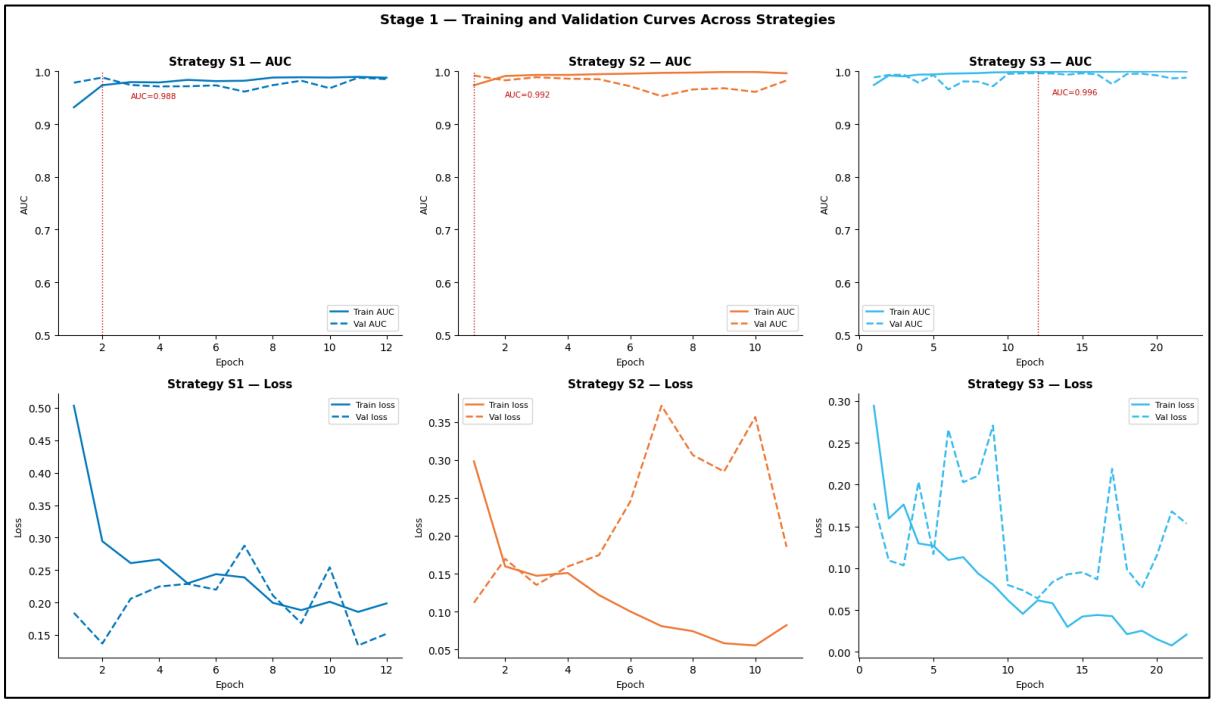


Figure 4.4 - Training and validation AUC (top) and loss (bottom) for all three strategies. Dotted lines mark the best validation epoch per strategy.

Strategy S1 (head only, 2.2% of parameters trained) achieved AUC 0.9848 over 12 epochs, demonstrating that ImageNet representations transfer meaningfully to chest radiograph classification without any convolutional adaptation. Strategy S2 unfroze Conv5_x alongside the head and pushed AUC to 0.9913 and specificity to 0.9367. It ran for 11 epochs, but best weights were already recorded at epoch 2, a clear sign of overfitting setting in early. Strategy S3 went further and trained everything. All 99.8% of parameters updated over 22 epochs, and the results showed it: AUC 0.9950, accuracy 0.9795, sensitivity 0.9844, specificity 0.9662, F1 0.9859, the highest across every metric. The sensitivity gain over S1 corresponds to approximately 6 additional correctly detected pneumonia cases per 641 on the validation set. Strategy S3 was selected for Stage 2.

4.4 Stage 2: CLAHE Preprocessing Evaluation

4.4.1 Primary Results

The S3 model was evaluated on the held-out test set (n = 879) under all four conditions with weights fixed throughout. Table 4.2 summarises results across all metrics and Figures 4.5 and 4.6 illustrate the metric comparison and ROC curves.

Condition	AUC	Accuracy	Sensitivity	Specificity	F1	FN	FP
C0	0.9979	0.9829	0.9875	0.9706	0.9883	8	7
C1	0.9892	0.9477	0.9626	0.9076	0.9641	24	22
C2	0.9809	0.9295	0.9392	0.9034	0.9510	39	23
C3	0.9931	0.9499	0.9423	0.9706	0.9649	37	7

Table 4.2 - Stage 2 test set performance across CLAHE conditions C0 to C3 (n = 879). (FN = false negatives. FP = false positives)

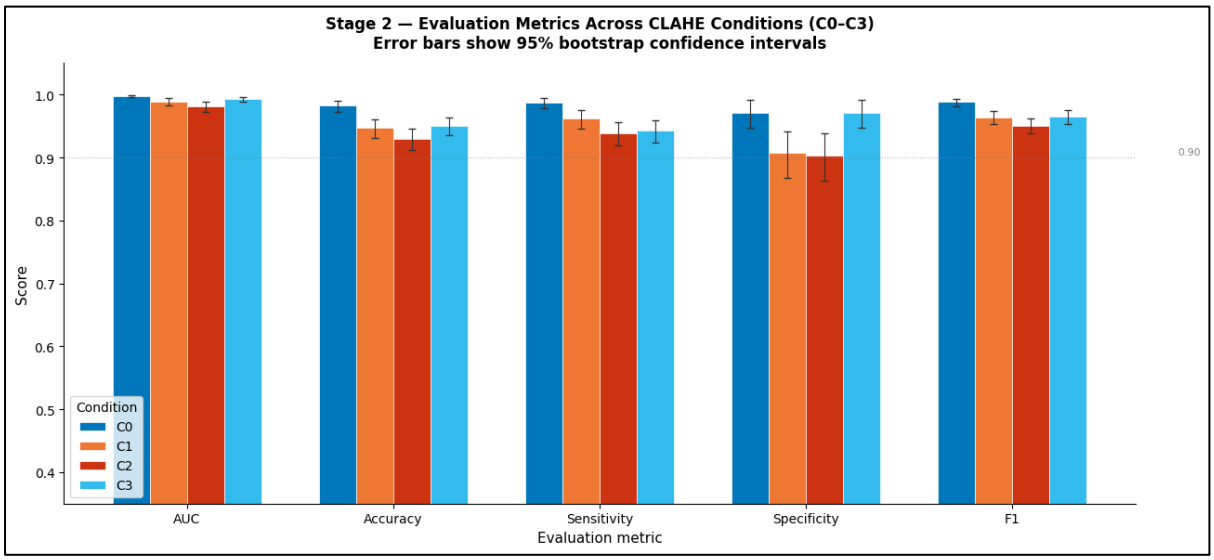


Figure 4.5 - All five evaluation metrics across conditions C0 to C3.

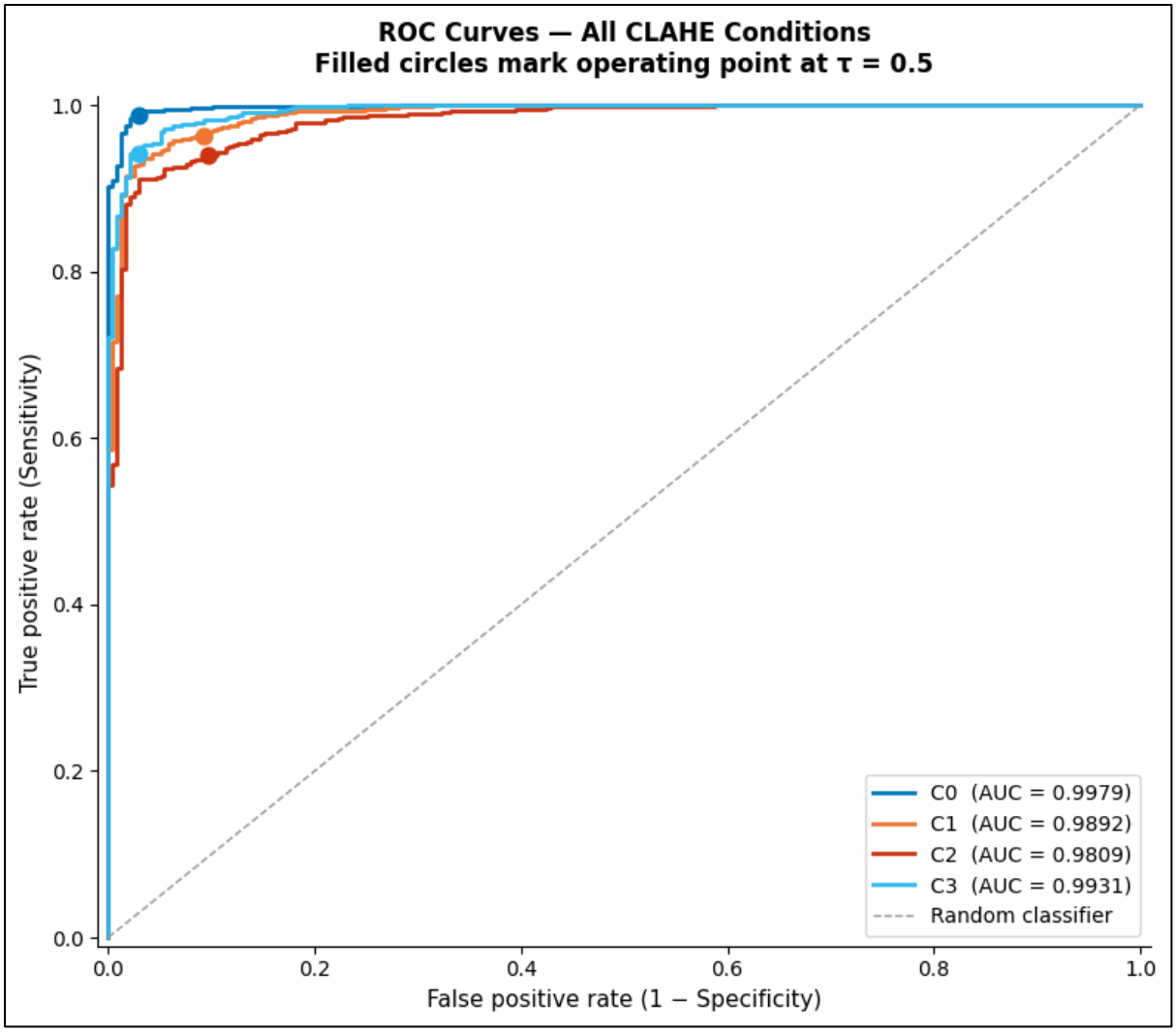


Figure 4.6 - ROC curves for all four conditions. Filled circles mark the operating point at τ = 0.5.

4.4.2 Condition Analysis

Condition C0 produced the strongest result across all metrics with AUC 0.9979, sensitivity 0.9875, specificity 0.9706, F1 0.9883 with only 15 total errors from 879 images. The model correctly detected 633 of 641 pneumonia cases (98.8%) and correctly identified 231 of 238 normal cases (97.1%), meeting all three pre-specified clinical thresholds (AUC ≥ 0.95, sensitivity ≥ 0.90, specificity ≥ 0.90) simultaneously. All three CLAHE conditions degraded performance across every metric relative to C0, with no improvement observed on any individual measure. Condition C1 (clipLimit = 2.0, tileGridSize = (8,8)) produced the most moderate degradation — AUC fell to 0.9892 and total errors rose from 15 to 46, with both false negatives (8 to 24) and false positives (7 to 22) increasing. Condition C2 was the worst of the four. A clip limit of 3.0 with the same (8,8) grid produced AUC 0.9809, sensitivity dropped to 0.9392, specificity to 0.9034, and the model racked up 62 total errors which was more than four times the baseline. Notably, C2 produced the largest false negative count (39 missed pneumonia cases) while C3 produced an equivalent false positive count to C0. Condition C3 (clipLimit = 2.0, tileGridSize = (4,4)) preserved specificity at 0.9706, matching C0 exactly, but sensitivity fell to 0.9423, with 37 false negatives compared to 8 under C0.

Confusion matrices are shown in Figure 4.7 and the sensitivity-specificity trade-off in Figure 4.8.

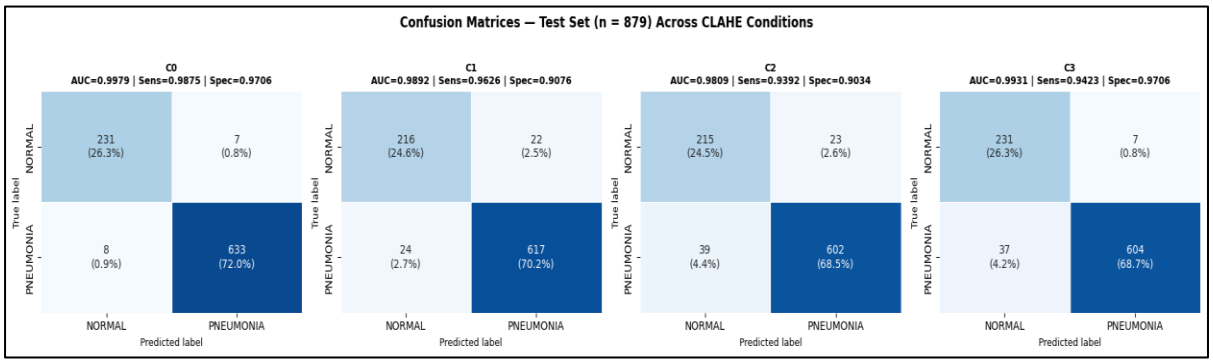


Figure 4.7

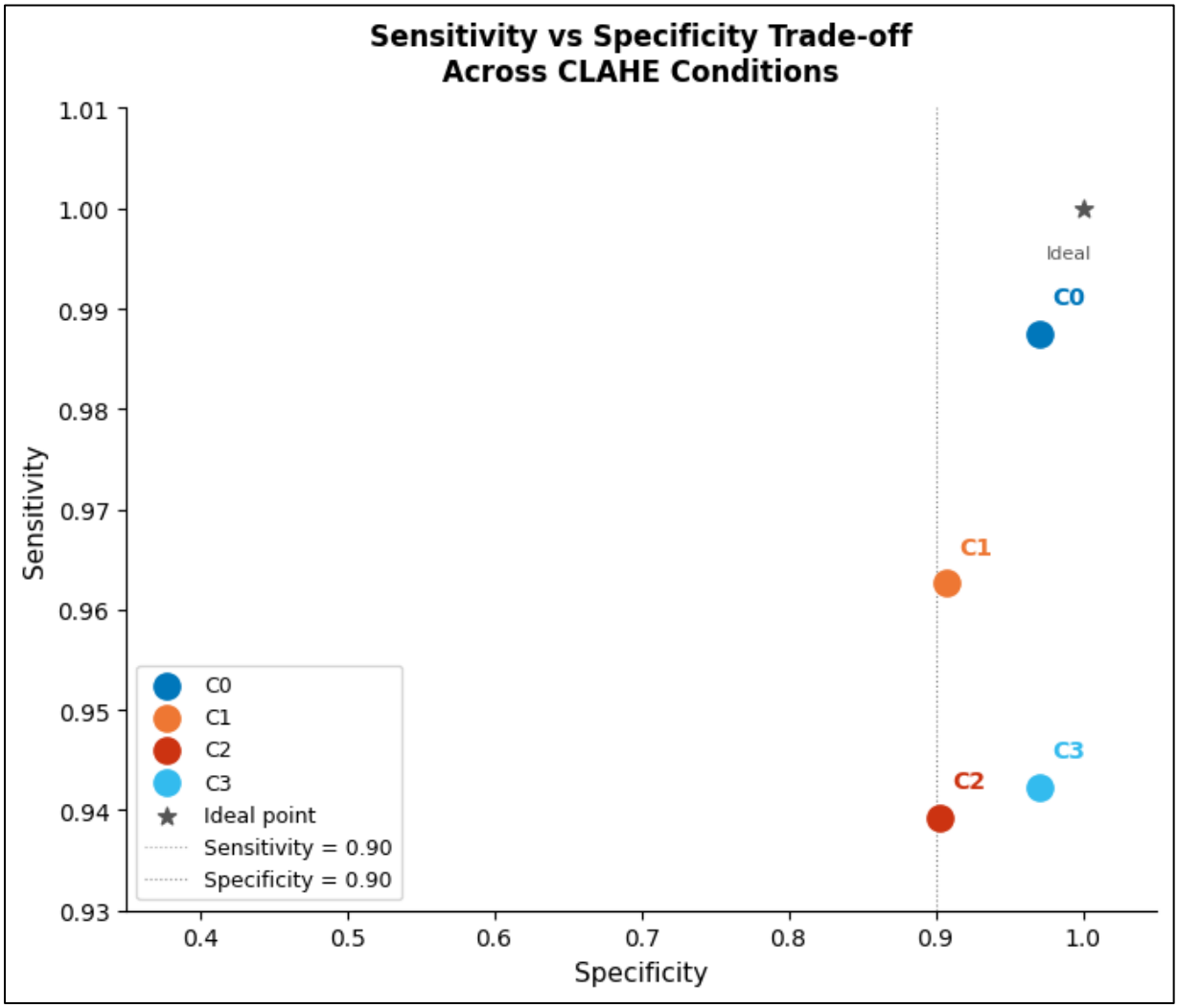


Figure 4.8

4.4.3 Statistical Testing

Bootstrap confidence intervals (95%, 1,000 resamples) and pairwise statistical tests are presented in Table 4.4. DeLong tests confirmed that C1 ($Z = 2.614$, $p = 0.009$) and C2 ($Z = 4.101$, $p < 0.001$) produced statistically significantly lower AUC than C0. C3 did not produce a statistically significant AUC difference relative to C0 ($Z = 1.532$, $p = 0.125$), though its point estimate was lower. McNemar tests revealed that all three CLAHE conditions produced significantly different error patterns relative to C0 (C1: $\chi^2 = 17.633$, $p < 0.001$; C2: $\chi^2 = 38.016$, $p < 0.001$; C3: $\chi^2 = 34.031$, $p < 0.001$), confirming that each CLAHE condition systematically changes which images are misclassified rather than simply adding random noise to the predictions. Parameter isolation comparisons confirmed clip limit as the dominant driver, increasing it from 2.0 to 3.0 significantly worsened AUC (C1 versus C2: $Z = 2.160$, $p = 0.031$), while tile grid size alone did not produce a statistically significant AUC difference (C1 versus C3: $Z = -1.217$, $p = 0.224$). Prediction confidence distributions are shown in Figure 4.9.

Comparison	Test	Statistic	p-value
C0 vs C1	DeLong	$Z = 2.614$	0.0090
C0 vs C2	DeLong	$Z = 4.101$	< 0.001
C0 vs C3	DeLong	$Z = 1.532$	0.1254
C0 vs C1	McNemar	$\chi^2 = 17.633$	< 0.001
C0 vs C2	McNemar	$\chi^2 = 38.016$	< 0.001
C0 vs C3	McNemar	$\chi^2 = 34.031$	< 0.001
C1 vs C2	DeLong	$Z = 2.160$	0.0308
C1 vs C3	DeLong	$Z = -1.217$	0.2237

Table 4.4 - Pairwise statistical comparisons. Significance threshold $\alpha = 0.05$.

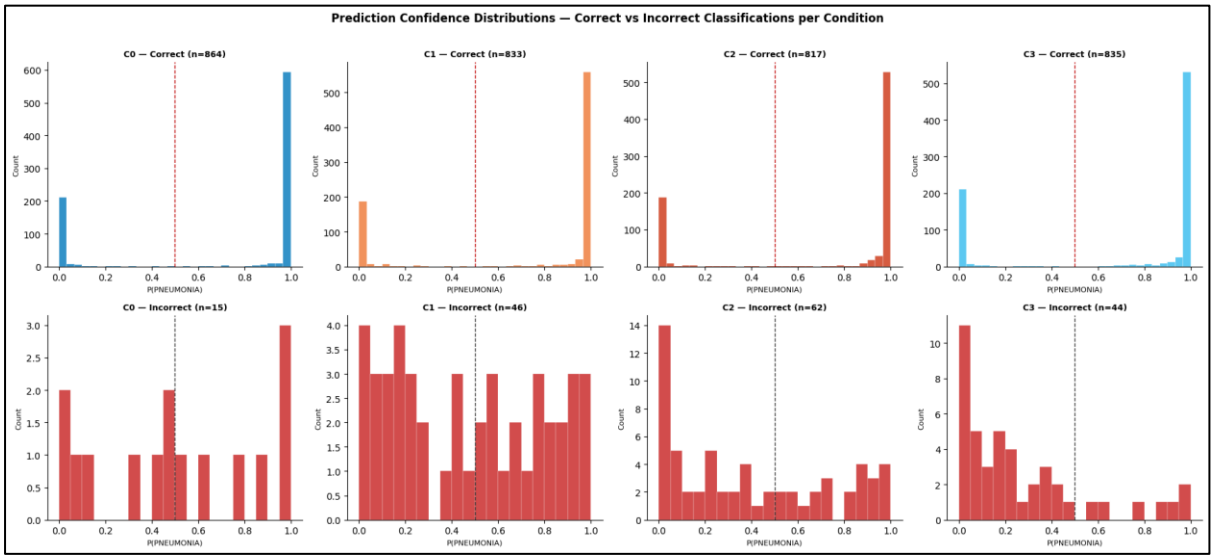


Figure 4.9

4.5 Misclassification and Grad-CAM Analysis

Under C0, the mean false negative confidence was 0.7543, indicating that the 8 missed cases were genuinely uncertain rather than high-certainty errors. Grad-CAM analysis of the

highest-confidence C0 false negatives (Figure 4.10, Figure 4.11) showed model attention concentrated at image peripheries rather than over the lung fields in most cases, indicating failure to engage with the anatomical region containing subtle consolidation rather than systematic misclassification. Across CLAHE conditions, the dominant pattern as summarised in Table 4.2, is a substantial increase in false negatives, with CLAHE at inference time primarily impairing the model's ability to detect pneumonia while leaving normal classification intact under the conservative C3 parameterisation. False positive Grad-CAM heatmaps showed near-complete gradient collapse under CLAHE conditions, with no identifiable focal activation region, indicating classification driven by global distributional features rather than specific anatomical findings.

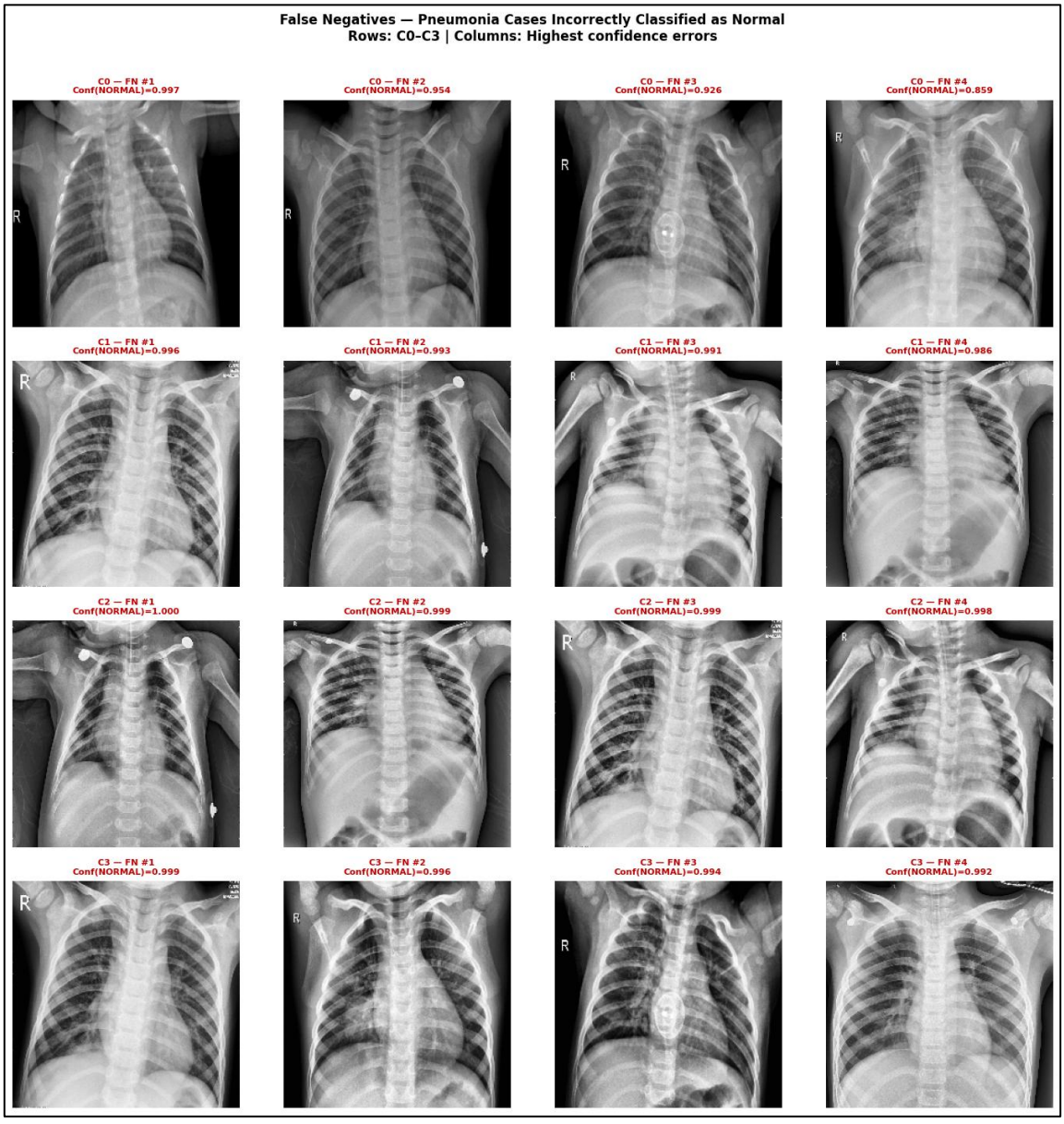


Figure 4.10 - Highest-confidence false negatives

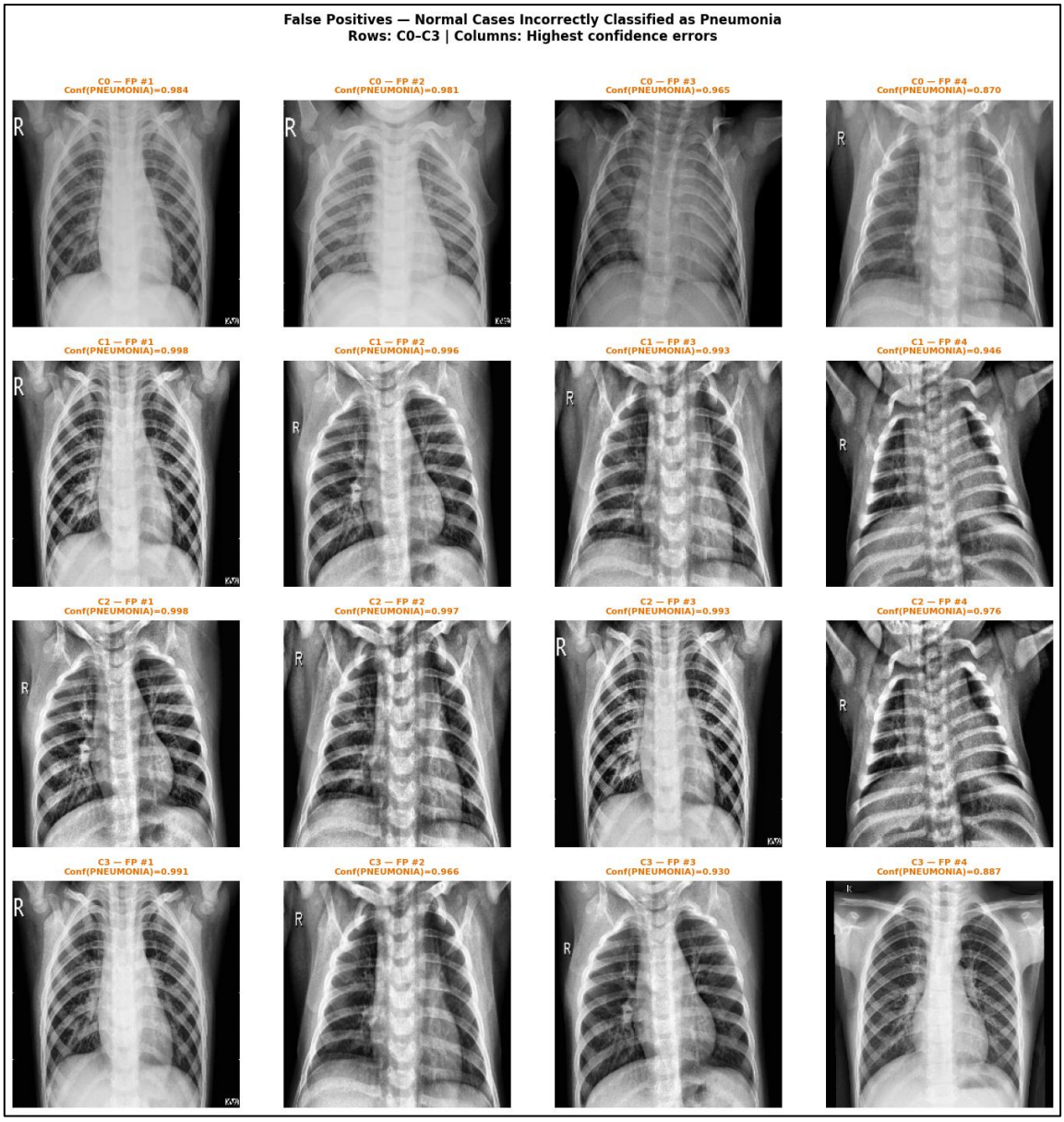


Figure 4.11 - Highest-confidence false positives

The most analytically significant Grad-CAM finding is shown in Figure 4.12, which tracks the same pneumonia image across all four conditions. Under C0, attention sits at image borders showing the model fails to attend to the lung fields. Under C1 and C3, attention patterns shift partially but remain insufficient to cross the decision threshold. Under C2, the attention becomes more diffuse. Grad-CAM on true positives (Figure 4.13, Fig. 4.14) confirmed coherent attention over the lung parenchyma under C0, validating that correct pneumonia detections were grounded in anatomically appropriate features. Under CLAHE conditions, true positive heatmaps became progressively more diffuse, suggesting that contrast enhancement even when the prediction is correct disrupts the spatial coherence of the model's feature activation.

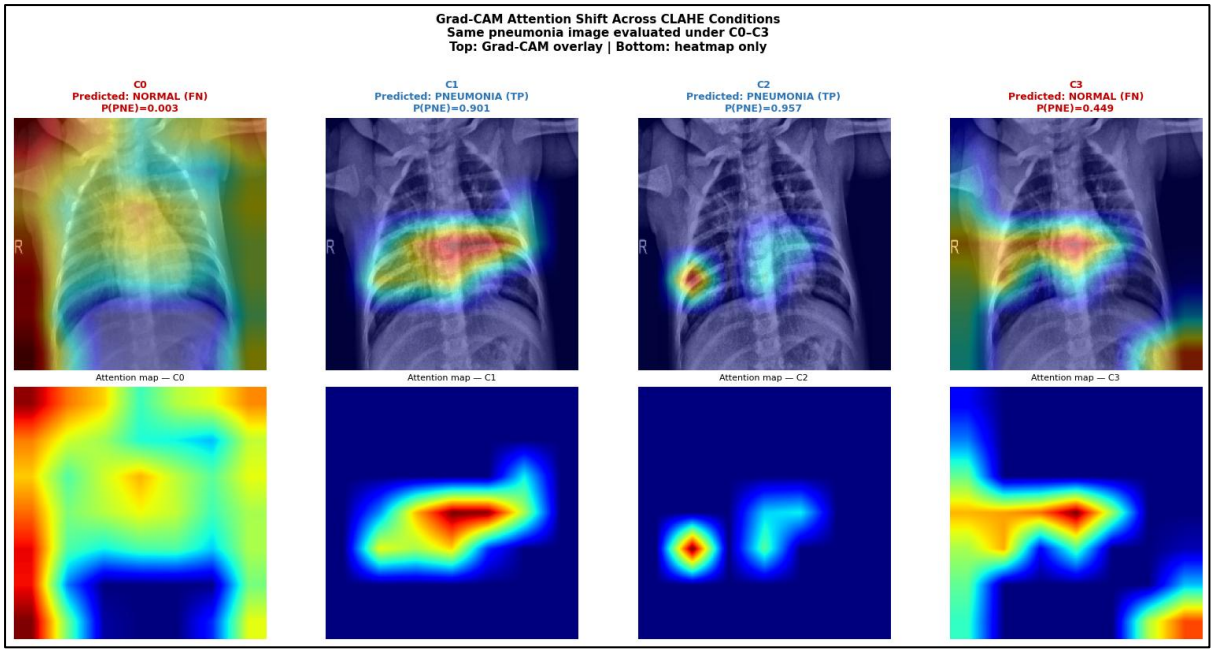


Figure 4.12 - Grad-CAM attention shift for the same pneumonia image across all four conditions, showing how preprocessing changes model spatial attention.

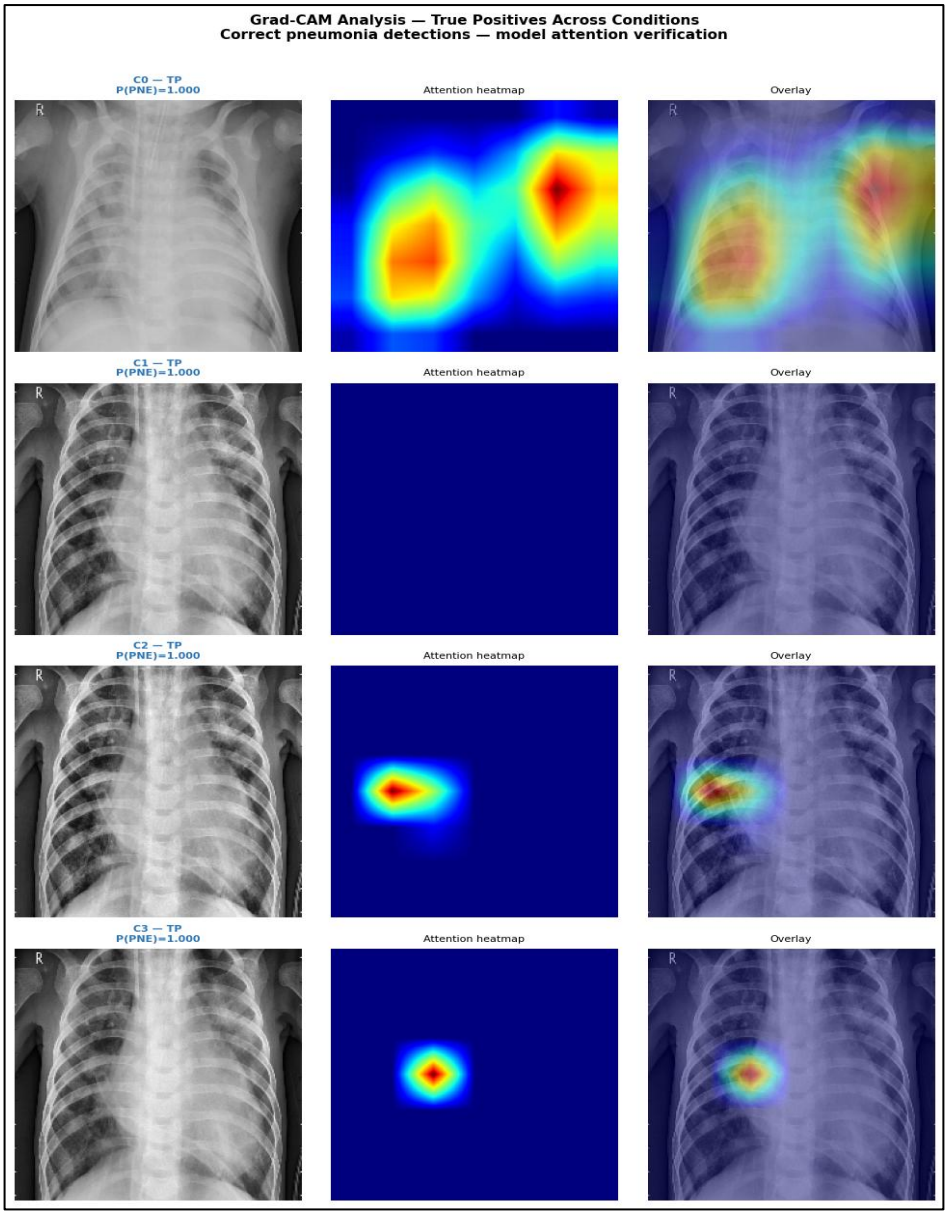


Figure 4.13 - Grad-CAM heatmaps for true positives

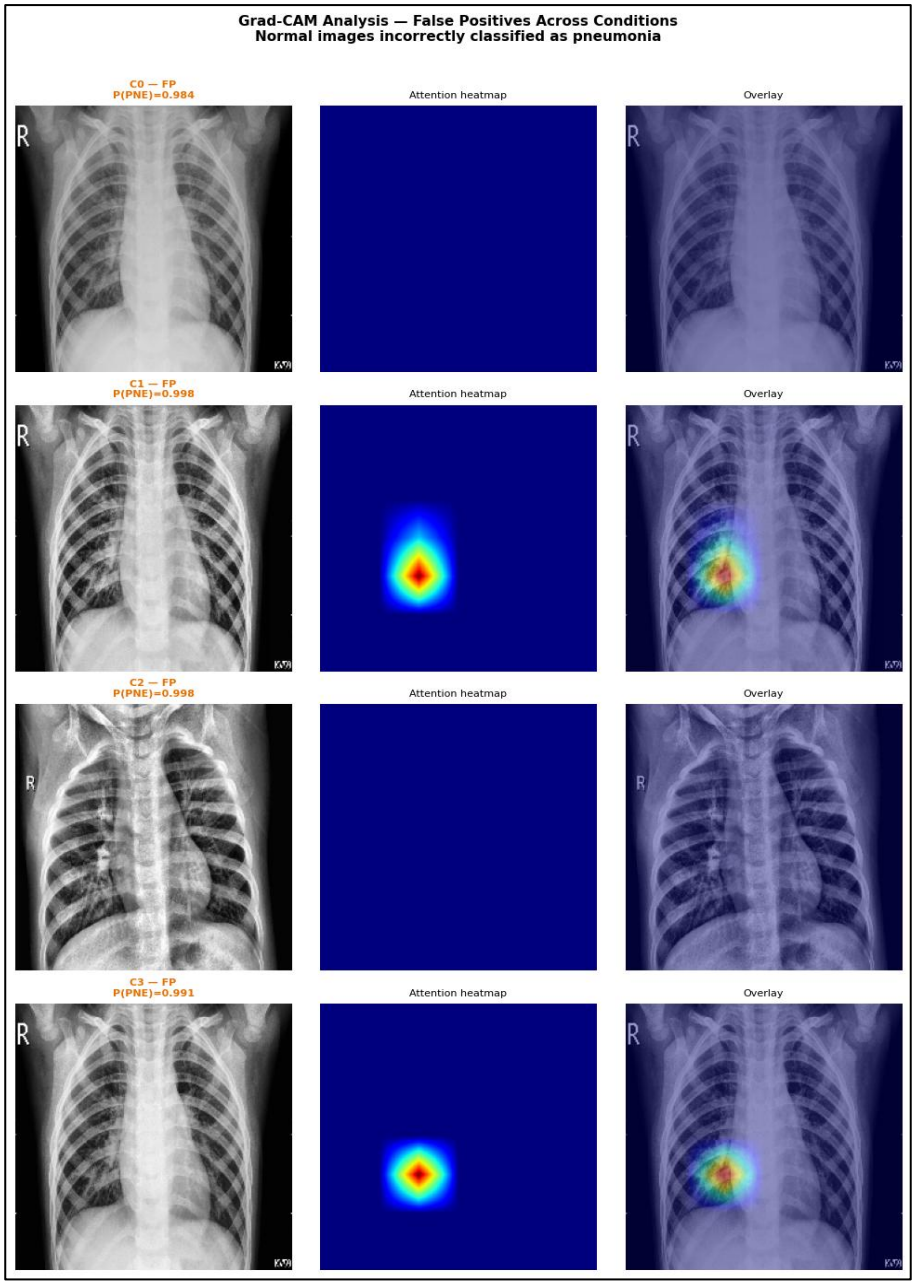


Figure 4.14 - Grad-CAM heatmaps for false positives

5. Discussion

5.1 Baseline Model Performance

The S3 model achieved AUC 0.9979, sensitivity 0.9875, and specificity 0.9706 under the baseline condition C0, meeting all three clinical thresholds simultaneously and with considerable margin (Section 4.4.2). These are strong results for binary classification on this benchmark. Correct predictions clustered at high confidence, consistent with stable, well-separated class representations. The mean false negative confidence of 0.7543 confirmed that missed cases were genuinely uncertain which is a more clinically manageable failure mode than systematic high-confidence misclassification. Grad-CAM analysis (Section 4.5) showed that failure in false negative cases was driven by failure to attend to the relevant anatomical region rather than a systematic classification bias (Walker et al., 2014).

Comparing against the existing literature, 0.9979 AUC holds up well. Diana et al. (2025) got 89.12% test accuracy using ResNet50 with gamma correction and CLAHE combined though their problem was multi-class and pulled from three different hospital sites, so a direct comparison does not really hold up. Direct accuracy comparison is complicated by the difference in task complexity, but the directional distinction is the critical point: Diana et al. found CLAHE improved performance when applied consistently across training and evaluation, while the present study found it degraded performance when applied only at inference time. That distinction is the core finding of the study.

5.2 Effect of CLAHE at Inference Time

CLAHE applied exclusively at inference time did not improve performance under any of the three conditions tested. Every condition degraded every metric relative to C0 — there was no condition under which inference-time CLAHE produced a net benefit on any individual measure. The character of degradation varied meaningfully across parameterisations. Conditions C1 and C2 produced statistically significantly lower AUC than C0 (DeLong: C1 $Z = 2.614$, $p = 0.009$; C2 $Z = 4.101$, $p < 0.001$). Condition C3 did not produce a statistically significant AUC difference relative to C0 ($Z = 1.532$, $p = 0.125$), though its point estimate was lower. All three CLAHE conditions produced significantly different error patterns from C0 by McNemar testing (C1 $\chi^2 = 17.633$; C2 $\chi^2 = 38.016$; C3 $\chi^2 = 34.031$; all $p < 0.001$), confirming that each condition systematically changes which images are misclassified.

As shown in Table 4.2, the dominant degradation pattern is a substantial increase in false negatives rather than false positive inflation. False positives remained relatively stable across all conditions, with C3 matching C0 exactly at 7. This means CLAHE at inference time primarily impairs the model's ability to detect pneumonia while leaving normal classification largely intact under conservative parameterisation. Condition C3 is the most analytically interesting: it preserved specificity exactly at 0.9706 which is identical to C0 while sensitivity fell from 0.9875 to 0.9423. Condition C2 was the worst performing overall, producing the largest false negative count and the lowest AUC of 0.9809. As confirmed by the parameter isolation comparisons in Section 4.4.3, clip limit was the dominant CLAHE parameter — tile grid size alone did not produce a statistically significant AUC difference.

5.3 Mechanistic Explanation

5.3.1 Training-Test Distribution Mismatch

The performance degradation across all three CLAHE conditions is explained by training-test distribution mismatch. The model was trained on unprocessed images. CLAHE shifts the pixel intensity distribution of test images away from the training distribution, altering the effective decision boundary without any retraining to compensate. The boundary has not moved but the data has. The exploratory analysis established that 18.5% of pneumonia training images fell below RMS contrast 0.35, compared to 2.5% of normal images which is a sevenfold difference identifying a substantial low-contrast subset where contrast enhancement changes the input most dramatically. CLAHE preferentially shifts these low-contrast images, creating a mismatch between the enhanced test distribution and the unprocessed training distribution the model was calibrated against. The result is that the model's learned decision boundary is no longer correctly positioned relative to the modified inputs, producing systematic errors that are consistent with distributional displacement rather than random noise.

The specific pattern of increased false negatives rather than false positives is consistent with this mechanism. If CLAHE shifts pneumonia images away from the high-intensity, high-contrast profile the model learned to associate with the positive class during training, those images become harder for the model to classify correctly not because the pathology is absent, but because the enhanced appearance no longer matches the learned representation. The primary risk of inference-time CLAHE on this model is missed pneumonia diagnoses.

5.3.2 Grad-CAM Evidence

Grad-CAM provides direct spatial confirmation of the distributional mismatch mechanism. Under C0, true positive detections showed coherent attention over the lung parenchyma, confirming the model responds to clinically relevant radiographic features when operating on the training distribution. Under CLAHE conditions, false positive heatmaps showed near-complete gradient collapse with no identifiable focal activation region with high-confidence pneumonia predictions on normal images with no anatomical basis, indicating classification driven by global distributional features rather than specific pneumonia findings. Tjoa and Guan (2020) identified the need for spatial validation as a complement to aggregate metrics precisely because models can achieve strong quantitative results through non-clinical feature associations. The Grad-CAM evidence here provides the spatial grounding that the aggregate metrics alone cannot supply.

The anchor image analysis (Figure 4.11) illustrates how CLAHE changes model attention on the same pneumonia image across all four conditions. The attention patterns shift with each condition, but the critical observation is that the shift does not become more anatomically specific under CLAHE rather, it becomes less coherent. Under C0, attention was at image borders, indicating failure to engage with the lung fields. Under CLAHE conditions, attention became progressively more diffuse rather than more focused on the relevant anatomical

region. This is inconsistent with CLAHE genuinely improving pneumonia feature detection and consistent with a distributional shift that changes what the model responds to without making that response more clinically grounded.

5.4 Implications

The results establish a clear boundary condition on the applicability of inference-time CLAHE. The mechanism that makes it effective in training-time preprocessing studies is absent when enhancement is applied only at inference time. What remains is only the distributional shift, which the model cannot compensate for because it has never seen enhanced images during training. The practical implication is direct: a clinician or system administrator cannot improve AI-assisted pneumonia detection by applying CLAHE to incoming images at deployment time if the model was trained without it. The preprocessing that was intended to help the model find subtle pneumonia features instead moves the input distribution away from the one the model was calibrated against, producing missed diagnoses rather than the intended sensitivity improvement. This is not an argument against CLAHE as a technique and, the literature consistently supports it when applied consistently across training and evaluation. The present study adds a specific and previously untested boundary condition: the benefit of CLAHE requires the model to have been trained on enhanced images.

5.5 Limitations

The most significant limitation is that all experiments were conducted on a single dataset from a single institution. The magnitude and character of the CLAHE-induced performance degradation is specific to the contrast properties of the Kermany et al. (2018) dataset, and a dataset with different baseline characteristics might produce different results under the same parameterisations. This limitation is intrinsic to the experimental design introducing a second dataset would have added confounds that prevent clean attribution of performance differences to the preprocessing condition, trading internal validity for external validity. The findings should be interpreted as a controlled proof-of-concept rather than a definitive deployment evaluation. Additionally, all binary metrics were computed at a fixed decision threshold of $\tau = 0.5$. In clinical deployment, threshold optimisation would be standard practice and may produce a different sensitivity-specificity balance under CLAHE conditions. The stratified split was performed at image rather than patient level due to the absence of patient identifiers in the Kaggle release, introducing a small risk of data leakage that cannot be fully mitigated without access to the original patient-level metadata.

6. Conclusion

6.1 Summary of Main Results

This study investigated whether CLAHE contrast enhancement applied exclusively at inference time improves ResNet50 pneumonia detection performance, and how the effect varies across different CLAHE parameterisations. The answer to both parts of the research question is clear.

CLAHE preprocessing applied at inference time did not improve performance under any of the three conditions tested. The baseline condition C0 produced the strongest result across all five-evaluation metrics: AUC 0.9979, sensitivity 0.9875, specificity 0.9706, accuracy 0.9829, and F1 0.9883 on the held-out test set of 879 images. All three pre-specified clinical thresholds were met simultaneously. All three CLAHE conditions degraded all metrics relative to C0. Conditions C1 and C2 produced statistically significantly lower AUC than C0 (DeLong, both $p < 0.05$). All three produced significantly different error patterns from C0 (McNemar, all $p < 0.001$). The dominant degradation pattern was a substantial increase in false negatives rather than false positive inflation, with C3 notably preserving specificity at the C0 level while sensitivity fell substantially (Section 4.4.2, Table 4.2).

The primary mechanism is training-test distribution mismatch. CLAHE shifted the intensity distribution of test images away from the training distribution, altering the effective decision boundary without retraining. The result was that CLAHE-processed images became harder for the model to classify correctly because their enhanced appearance no longer matched the learned representation, producing increased missed diagnoses rather than the intended sensitivity improvement. Grad-CAM gradient collapse on CLAHE false positive cases provided direct spatial confirmation of this mechanism. The full statistical evidence and mechanistic explanation are presented in Sections 4.4.3 and 5.3.

6.2 Appraisal of Original Aims

The study was structured around six objectives stated in Chapter 1. Each is appraised below.

1. The full 5,856-image dataset was downloaded, audited for integrity, and partitioned using a stratified split that preserved the 73:27 pneumonia-to-normal ratio to within ±0.1% across all three partitions, confirmed by a hard assertion. A seven-image discrepancy against the published total of 5,863 was documented as a methodological limitation.

2. Mean pixel intensity, standard deviation, and RMS contrast were computed on a stratified training sample of 200 images per class. A statistically significant difference in pixel intensity standard deviation was confirmed (Mann-Whitney $U = 27{,}531$, $p < 0.001$). The low-contrast tail analysis found that 18.5% of pneumonia images fell below RMS contrast 0.35, compared to 2.5% of normal images, providing direct quantitative motivation for CLAHE as a preprocessing intervention.

3. Three strategies were implemented and trained from identical ImageNet-pretrained weights with seeds reset before each build. Strategy S3 (full fine-tuning, 99.8% of parameters updated) achieved the highest validation AUC of 0.9950 and was selected for Stage 2. The

strategy comparison confirmed empirically that full fine-tuning outperforms both head-only and partial unfreezing on this dataset.

4. The S3 model was evaluated on the full 879-image test set under conditions C0 through C3 with weights fixed throughout. All five metrics were computed for each condition alongside per-class precision and recall, and raw prediction probabilities were saved for downstream statistical analysis.

5. Bootstrap confidence intervals were computed over 1,000 resamples for all five metrics. DeLong tests were applied to all pairwise AUC comparisons including parameter isolation comparisons between CLAHE conditions. McNemar tests with continuity correction were applied to all pairwise classification outcome comparisons. All tests were implemented from first principles, saved to CSV, and reported in Section 4.4.3.

6. Grad-CAM heatmaps were generated for true positive, false negative, and false positive cases under each condition, targeting the final Conv5_x convolutional layer. Four figures were produced covering C0 false negative analysis, condition-comparative analysis of the same image across all four conditions, true positive attention verification, and false positive gradient collapse analysis. Grad-CAM gradient collapse on CLAHE false positive cases provided the key mechanistic evidence connecting the statistical findings to the distributional mismatch explanation.

6.3 Future Work

6.3.1 Training on CLAHE-Enhanced Images

The most important extension is to train a ResNet50 model on CLAHE-enhanced images and evaluate it under matching conditions, directly eliminating the distributional asymmetry identified as the failure mechanism. A fully factorial design with no CLAHE, training only, inference only, and both would cleanly isolate the contribution of each configuration and produce a complete picture of how CLAHE interacts with the training pipeline. Based on Diana et al. (2025), consistent application is expected to produce performance at least equal to C0 and potentially above it, particularly for the 18.5% low-contrast pneumonia subset identified in the exploratory analysis.

6.3.2 Cross-Site Validation

The experimental design used a single dataset, which controlled for confounds but limited external validity. Evaluating the trained model on a second dataset from a different institution such as the RSNA Pneumonia Detection Challenge collection would provide a direct test of whether inference-time CLAHE offers any benefit in the context of genuine cross-site distributional shift, where the mismatch is caused by real differences in scanner hardware and acquisition protocols rather than by the preprocessing itself.

6.3.3 Alternative Preprocessing Approaches

Histogram matching aligns the intensity distribution of incoming test images to a reference training distribution, directly targeting the distributional mismatch mechanism identified here by design. It represents a conceptually more appropriate inference-time intervention than

CLAHE for the deployment problem investigated and would be a natural next comparison in the same experimental framework. Threshold optimisation selecting τ to maximise a clinical utility function rather than fixing it at 0.5 would also provide a more complete picture of the operating characteristics of each CLAHE condition and may identify parameterisations that are clinically acceptable at an adjusted threshold.

The finding establishes a clear boundary condition: the benefit of CLAHE as a preprocessing intervention is contingent on the model having been trained on enhanced images. Applying it only at inference time reverses the expected effect. The most important direction for future work is to test CLAHE applied consistently across training and evaluation, which would directly address the identified mechanism and determine whether the technique can deliver the sensitivity improvement it was hypothesised to provide for the low-contrast pneumonia subset that motivates this line of investigation.

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Appendix A: Poster

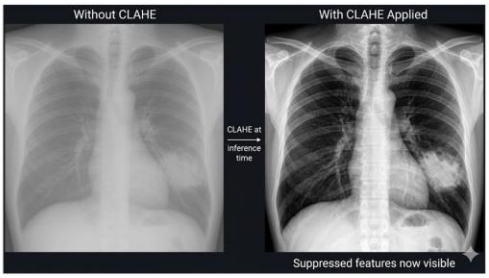
Evaluating the Effect of Inference-Time CLAHE Contrast Enhancement on ResNet50 Pneumonia Detection: A Controlled Experimental Study

BSc (Hons) Computer Science
Faculty of Science and Engineering
10 March 2026

a.r.u. | Anglia Ruskin University

RESEARCH PROBLEM

Deep learning models for pneumonia detection achieve strong performance when training and test data share the same preprocessing pipeline. In deployment, however, models are often fixed and cannot be retrained, yet incoming images may arrive with different contrast characteristics from the training distribution. CLAHE contrast enhancement has been shown to improve performance when applied consistently across training and evaluation (Diana et al., 2025), but its effect when applied exclusively at inference time to a model trained on unprocessed images has never been systematically tested.

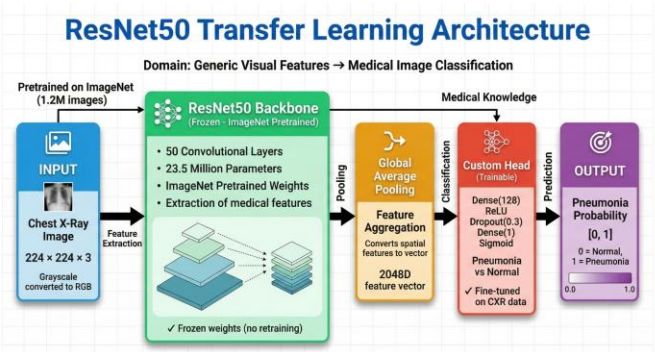


AIM

To evaluate whether CLAHE contrast enhancement applied exclusively at inference time improves ResNet50 pneumonia detection performance on the Kermany et al. (2018) dataset, and to determine how performance varies across different CLAHE parameterisations using a controlled two-stage experimental pipeline.

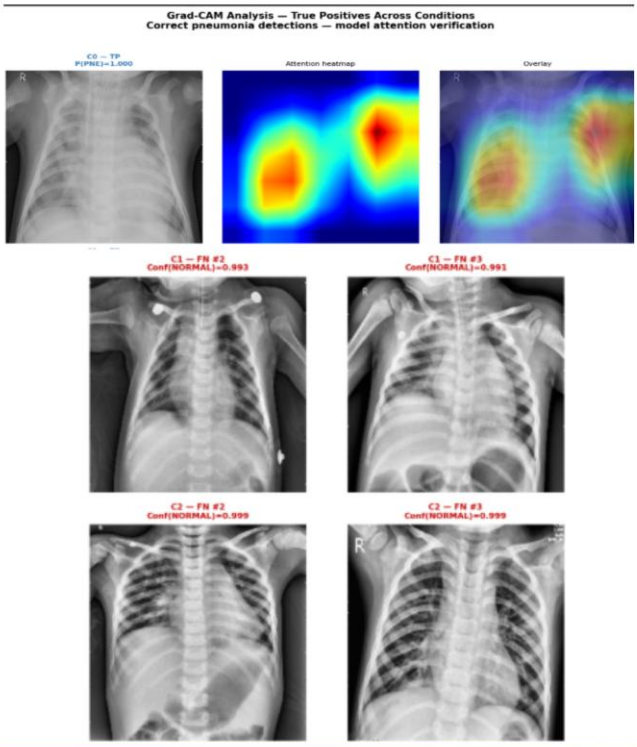
METHODOLOGY

- The Kermany et al. (2018) chest X-ray dataset (5,856 images) was used with a 73:27 pneumonia-to-normal class ratio.
- Three ResNet50 fine-tuning strategies (head only, partial unfreezing, full fine-tune) were evaluated on the validation set.
- Four CLAHE preprocessing conditions were applied exclusively at inference time to the fixed S3 model.
- Performance evaluated using AUC, sensitivity, specificity, F1, and accuracy with bootstrap confidence intervals, DeLong AUC tests, and McNemar paired outcome tests.
- Grad-CAM applied to true positive, false negative, and false positive cases under all four conditions to provide mechanistic spatial evidence for the observed performance patterns.



PRELIMINARY RESULTS

CLAHE preprocessing applied exclusively at inference time degraded performance under all three experimental conditions relative to the unprocessed baseline (C0). Grad-CAM gradient collapse on false positive cases under CLAHE conditions confirmed training-test distribution mismatch as the primary mechanism: classifications were driven by global distributional features rather than anatomical pneumonia findings.



REFERENCES

Diana, D., Ermatita, E. and Abdiansah, A. (2025) 'Enhanced chest X-ray image classification using deep learning with CLAHE and gamma correction', 2025 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS), IEEE, pp. 108–113.

Poster by - Aleta Shibu
SID - 2173844
Supervisor - Prof. Hossein Abroshan

Appendix B: Interim Report

1. Project Proposal

Project Title:

Evaluating the Effect of Inference-Time CLAHE Contrast Enhancement on ResNet50 Pneumonia Detection: A Controlled Experimental Study

Aim:

The aim of this project is to design, implement, and evaluate a controlled experimental pipeline that assesses whether CLAHE contrast enhancement applied exclusively at inference time improves ResNet50 pneumonia detection performance on the Kermany et al. (2018) chest X-ray dataset, and how the effect varies across different CLAHE parameterisations.

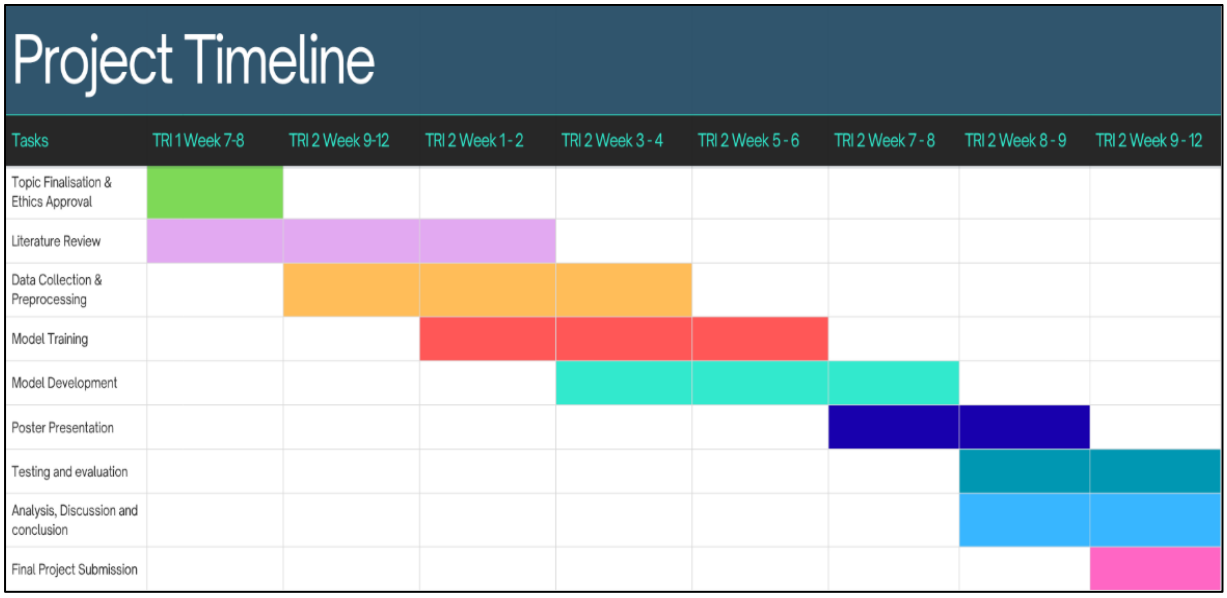
Objectives:

- Obtain and preprocess the Kermany et al. (2018) chest X-ray dataset, constructing a stratified 70/15/15 train, validation, and test partition that preserves the natural 73:27 class distribution.
- Conduct exploratory pixel-level analysis to characterise the baseline intensity and contrast properties of normal and pneumonia images and quantify the proportion of low-contrast cases that motivate CLAHE as a preprocessing intervention.
- Fine-tune a ResNet50 architecture, pretrained on ImageNet, on the unprocessed training subset using three fine-tuning strategies with head only, partial unfreezing, and full network fine-tuning selecting the best configuration by validation AUC.
- Evaluate the trained model on the held-out test set under four CLAHE conditions: a baseline with no preprocessing and three parameterised conditions varying clip limit and tile grid size, computing AUC, sensitivity, specificity, F1, and accuracy per condition.
- Apply bootstrap confidence intervals, DeLong AUC comparison tests, and McNemar paired outcome tests to confirm the statistical significance of observed performance differences between conditions.
- Apply Grad-CAM to representative true positive, false negative, and false positive cases under each condition to provide mechanistic visual evidence for the observed performance patterns.

The project is implemented as a reproducible fourteen-block Jupyter notebook on Google Colab Pro using Python 3.10, TensorFlow 2.16.1, and OpenCV. The experimental design follows a two-stage pipeline with model selection on the validation set followed by fixed-model CLAHE evaluation on the held-out test set to prevent confounding between architecture selection effects and preprocessing effects. Results are evaluated against pre-specified clinical performance thresholds of AUC ≥ 0.95, sensitivity ≥ 0.90, and specificity ≥ 0.90.

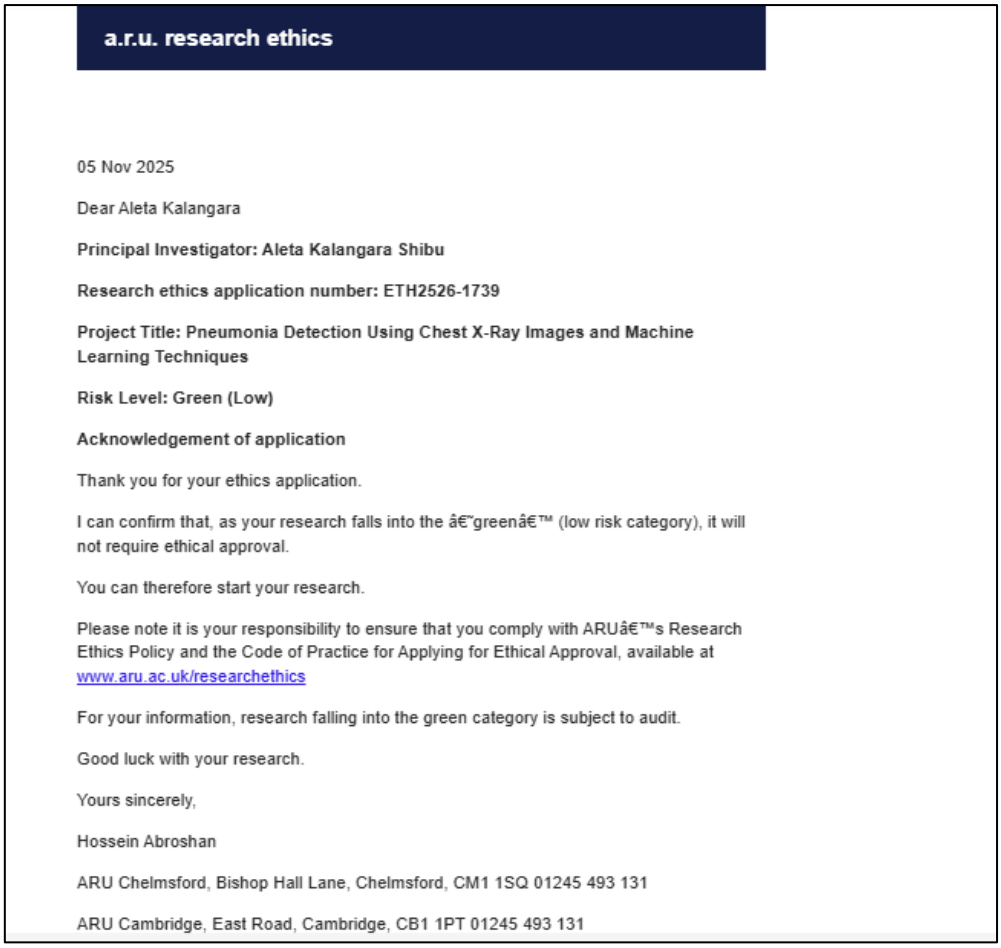
2. Work Plan / Gantt Chart

The timeline below covers the full project period from September 2025 to August 2026.



3. Confirmation of Ethics Approval

The project was submitted for ethical review through the ARU ethics application system. The study uses only publicly available, fully anonymised chest X-ray data from the Kermany et al. (2018) dataset, hosted on Kaggle. No personally identifiable information, human participants, or primary data collection is involved. The ethics application was approved. A screenshot of the confirmation email is included below.



a.r.u. research ethics

05 Nov 2025

Dear Aleta Kalangara

Principal Investigator: Aleta Kalangara Shibu

Research ethics application number: ETH2526-1739

Project Title: Pneumonia Detection Using Chest X-Ray Images and Machine Learning Techniques

Risk Level: Green (Low)

Acknowledgement of application

Thank you for your ethics application.

I can confirm that, as your research falls into the â€˜greenâ€™ (low risk category), it will not require ethical approval.

You can therefore start your research.

Please note it is your responsibility to ensure that you comply with ARUâ€™s Research Ethics Policy and the Code of Practice for Applying for Ethical Approval, available at www.aru.ac.uk/researchethics

For your information, research falling into the green category is subject to audit.

Good luck with your research.

Yours sincerely,

Hossein Abroshan

ARU Chelmsford, Bishop Hall Lane, Chelmsford, CM1 1SQ 01245 493 131

ARU Cambridge, East Road, Cambridge, CB1 1PT 01245 493 131

4. CV & Exit Plan

A copy of my current CV is included below, followed by my exit plan.

Aleta Shibu

Cambridge, UK · LinkedIn URL · Github URL

PERSONAL STATEMENT

I am a final-year Computer Science student at Anglia Ruskin University (predicted first class) with a specific research interest in trustworthy medical AI, bringing together rigorous machine learning methodology and real-world software engineering to create systems that clinicians can rely on. My final year project examines robustness under contrast-variation domain shift in medical image classification, and my hands-on experience building CI/CD pipelines and containerised infrastructure with Terraform and Docker has given me a concrete understanding of what robust deployment requires.

EDUCATION

BSc (Hons) Computer Science | Anglia Ruskin University, Cambridge | **Sept 2022 – May 2026 (Expected)**

Predicted First Class Honours

Key modules: Artificial Intelligence · Image Processing · Data Structures & Algorithms · Software Engineering · Cloud Computing · Computer Graphics Programming

Final year project (In progress): Pneumonia detection on chest x-rays under contrast & intensity shift: benchmarking transfer learning CNNs

- Benchmarking ResNet50, DenseNet, and EfficientNet under controlled contrast and intensity shift, evaluating how classification performance and calibration degrade as shift severity increases.
- Comparing image processing pipelines like Contrast Limited Adaptive Histogram Equalisation, gamma correction, and histogram matching to assess whether clinical image enhancement improves or harms robustness under distribution shift.

OTHER RELEVANT COURSEWORK

Multi-Modal AI Coursework (*Artificial Intelligence Module - Year 3)*

- Designed and evaluated machine learning pipelines across four distinct data modalities - tabular, image, audio, and text. Applied appropriate preprocessing, classification, and interpretability techniques to each, including SHAP, LIME, and attention-rollout.
- Maintained consistent evaluation methodology throughout, critically reflecting on results.

NLP Sentiment Analysis on Customer Reviews (*Research Methodologies Module - Year 2)*

- Applied Aspect-Based Sentiment Analysis (ABSA) with TF-IDF (text feature extraction) preprocessing to iPhone customer reviews, demonstrating how unstructured text can be transformed into structured, business-actionable insights identifying specific product aspects driving satisfaction and concern, and correlating sentiment polarity with real customer satisfaction scores.

RELEVANT PROFESSIONAL EXPERIENCE

Software Engineer (Year in Industry, Part-Time) | Office for National Statistics | **Sept 2024 – Present**

- Contributing as part of a cross-functional engineering team building and modernising survey infrastructure used by field interviewers and internal researchers across the Office for National Statistics Social Surveys division.
- Participating actively in agile ceremonies including daily stand-ups, sprint planning, and retrospectives.
- Resolving support tickets alongside planned sprint work as part of regular business-as-usual delivery.
- Collaborating closely with teammates through code review, pull request feedback, and pair programming sessions to maintain code quality and share technical knowledge across the team.
- Contributing to the migration of the Postcode Address File from legacy survey management systems to modern cloud infrastructure spanning Google Cloud backend migration, React user interface development and integrating TypeScript and Node.js to reduce manual operational processes.

- Building and maintaining continuous integration and deployment pipelines to support safer, faster deployments across survey tooling, including containerised database environments using infrastructure-as-code and Docker, enabling automated database testing.
- Developed and improved database stored procedures to automate operational survey workflows including transfer detection and identifier assignment, supported by full testing and deployment pipelines.
- Maintaining technical documentation standards to support team onboarding and knowledge transfer.

Employability Activator (Part-Time) | Anglia Ruskin University | **Sept 2025 – Present**

- Supporting the university's employability service through a peer-facing role aimed at encouraging students to actively engage with career development support and resources available to them.
- Representing the employability service at university events through information stands, creating accessible entry points for students.
- Contributing to digital promotion efforts through social media content and blogs to raise awareness of employability resources and broaden reach across the student community.

SKILLS

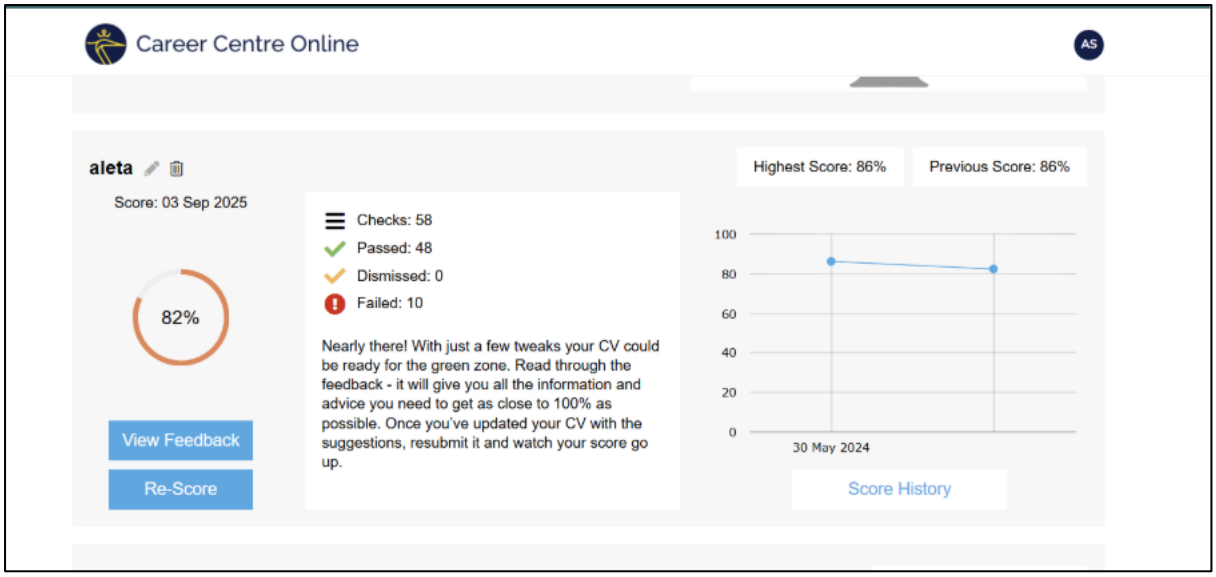
Coding Languages	Python · SQL · TypeScript · JavaScript · Node.js · C# · Visual Basic 6 (legacy)
Machine Learning	Supervised & unsupervised learning · Evaluation design (Precision/Recall/F1/ROC-AUC) · Model comparison
Software Engineering & DevOps	Continuous integration and deployment · Automated testing · Secure delivery · GitHub · Google Cloud · Docker · Terraform
Other	Collaboration · Communication · Problem solving · Adaptability · Attention to detail

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After completing my undergraduate degree, I would like to further develop my knowledge and technical expertise in the field of Artificial Intelligence. I plan to pursue a master's degree in Artificial Intelligence or Data Science, focusing on advanced topics such as deep learning, computer vision, and natural language processing.

5. Employability Activities

My CV was submitted to the ARU Careers Centre CV Checker and received a score of 80% or greater. A screenshot of the result is included below.


Career Centre Online
AS
aleta
Score: 03 Sep 2025
82%
View Feedback
Re-Score
Checks: 58
Passed: 48
Dismissed: 0
Failed: 10
Nearly there! With just a few tweaks your CV could be ready for the green zone. Read through the feedback - it will give you all the information and advice you need to get as close to 100% as possible. Once you've updated your CV with the suggestions, resubmit it and watch your score go up.
Highest Score: 86%
Previous Score: 86%
100
80
60
40
20
0
30 May 2024
Score History